

Rethinking the Validity of Physiological Signal-Based Mental Workload Assessment

Yuzheng Wang^a, Xiaodong Xu^a, Liang Ma^{a*}

^a*Department of Industrial Engineering, Tsinghua University, Beijing 100084, China*

Abstract

Mental workload assessment using physiological signals and machine learning has become common, with many studies reporting high classification accuracy. However, whether these approaches genuinely reflect human cognitive states remains unclear. This paper reviews current measurement methods and examines the physiological signal–machine learning pipeline through an information-theoretic lens, encompassing signal acquisition, preprocessing, feature extraction, labeling, model training, and evaluation. We identify several limitations: the many-to-many mapping between physiological signals and mental workload, information loss during preprocessing, the inadequacy of task-condition and self-report labels as proxies for experienced workload, poor model transferability across tasks and individuals, and the pervasive influence of inter- and intra-individual variability, emotional states, and expertise on physiological responses. We argue that current approaches often classify task conditions rather than mental workload itself, and that human variability is not noise to be eliminated but information to be incorporated. These limitations do not invalidate existing methods: under controlled conditions, physiological measures can still provide useful information. We propose a reconceptualization of the measurement problem and derive three directions for improvement: tightening the conditions under which current methods are applied, redesigning the measurement pipeline to preserve rather than discard uncertainty, and building models that adapt to rather than ignore human variability. We discuss continuous probabilistic estimation, temporal modeling, meta-learning for rapid personalization, online adaptation to baseline shifts, and disentangled representation learning for separating workload from emotional confounds. We conclude that although physiological signals contain information related to mental workload, future research should prioritize measurement validity, generalizability, and the quantification of uncertainty.

Keywords: Mental workload; Physiological signals; Machine learning; Construct validity; Human variability;

1. Introduction

Mental workload (MWL) is a key construct for characterizing the cognitive resources an individual expends during task execution. It has been widely adopted in human factors engineering and cognitive science, particularly in the design and evaluation of complex human–machine systems such as nuclear power plants, aviation cockpits, and driving systems (Charles & Nixon, 2019; Tierney et al., 2025). Suboptimal MWL—whether overload or underload—has been consistently linked to performance decrements, errors, and accidents in safety-critical settings (Young & Stanton, 2002). Accurate and reliable MWL assessment is therefore a

* Corresponding author. Tel.: +0-000-000-0000 ; fax: +0-000-000-0000 .
E-mail address: liangma@tsinghua.edu.cn.

practical necessity for enhancing system safety and individual well-being. Drawing on multiple resource theory and limited resource theory, MWL can be defined as the degree of activation of a finite pool of cognitive resources, mediated by external environmental factors and internal individual characteristics (Longo et al., 2022).

Physiological measures enable continuous, unobtrusive data collection without interfering with primary task performance. Common physiological modalities for MWL assessment include cardiovascular measures (e.g., heart rate and heart rate variability), respiratory and electrodermal measures (e.g., respiratory rate and skin conductance), oculomotor measures (e.g., pupil diameter and fixation behavior), and neurophysiological measures (e.g., electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS)) (Tierney et al., 2025).

In recent years, the advent of machine learning (ML) and deep learning (DL) has introduced a new paradigm for MWL assessment. The dominant approach follows a pipeline of physiological signal acquisition → feature extraction → labeling → model training → state prediction. Many studies have reported high classification accuracies, often exceeding 80% and occasionally 90%, fostering optimism about data-driven MWL monitoring (Cheema et al., 2018; Ding et al., 2020; Wilson & Russell, 2003).

Despite these promising results, several challenges remain insufficiently addressed. First, reported high accuracies may not replicate across experimental settings. Physiological responses to MWL are highly sensitive to task characteristics, environmental conditions, and individual differences (Ayres et al., 2021; Charles & Nixon, 2019). A classifier trained on one task—such as the mental arithmetic tasks used by Ding et al. (2020)—may not generalize to other tasks, such as simulated flight or nuclear power plant operations (Zhao et al., 2018). This generalization problem is compounded by domain specificity: cardiovascular measures proved more effective in driving studies than in nuclear power operations (Tao et al., 2019), suggesting that physiological responses to MWL are scenario-dependent rather than universal.

Second, even when a model performs well for a specific population, it may not transfer to others. Models trained on young, healthy student samples—the participant populations in Ding et al. (2020) and Moustafa et al. (2017)—may not generalize to trained professionals, older adults, or clinical populations. Such limited transferability across populations severely restricts the practical applicability of existing models.

Third, most supervised learning studies rely on task conditions (e.g., low vs. high difficulty) as proxies for ground-truth labels, assuming that task demand translates directly and deterministically into a specific MWL state. This assumption is problematic because MWL is a multidimensional experience mediated by individual differences, fatigue, coping strategies, and prior experience (Young et al., 2015). Two individuals performing the same task may experience vastly different levels of MWL, and the same individual may experience fluctuating MWL under constant task demand. Consequently, what many ML models learn may be task discrimination rather than MWL itself. This disconnect between the label (task condition) and the target construct (MWL) undermines the reliability and validity of existing models. Kumar Das et al. (2025) recently conducted a systematic review focused specifically on ground-truth labeling for cognitive workload adaptive systems, highlighting that the reliability and validity of labels remain a fundamental unresolved issue (Fairclough et al., 2005).

Fourth, current ML and DL approaches face additional interrelated problems: the non-unique mapping between physiological patterns and mental states, information loss during continuous signal sampling, the lack of standardized data-processing pipelines, the discrete versus continuous nature of MWL labeling, and the absence of independent validation standards for model effectiveness (Debie et al., 2021; Pereira et al., 2025).

Therefore, this paper does not aim to add yet another high-accuracy classifier to the literature. Instead, it addresses a more fundamental question: to what extent do physiological signals provide valid and reliable information about mental workload? Rather than pursuing incremental improvements within the existing framework, we critically examine the assumptions, limitations, and implicit biases of current approaches. Specifically, we investigate whether—and under what conditions—physiological signals can yield valid

measurements of MWL, trace methodological weaknesses across the signal-processing and machine-learning pipeline, and discuss potential directions for improvement. Our goal is not to discard machine learning but to realign its application with the theoretical and empirical realities of mental workload.

2. Physiological Measurement of Mental Workload

Building on multiple resource theory (Wickens, 2002) and limited resource theory (Kahneman, 1973), the working definition introduced in Section 1 by Longo et al. (2022) characterizes MWL as a multidimensional construct shaped by external environmental and situational factors as well as internal individual characteristics such as skills and past experience. Against this conceptual background, this section reviews the physiological measures commonly used for MWL assessment and the evidence linking each of them to workload.

2.1. Task Performance and Self-Report Measures

The validity of physiological measures is often assessed against task performance and self-report measures, which serve as proxy labels or reference standards (Charles & Nixon, 2019; Prasetyo & Iridiastadi, 2025).

Task performance measures assess an individual's performance on primary or secondary tasks; they reflect functional capacity but cannot distinguish the source of workload variations. Self-report measures provide subjective workload ratings, typically administered post-task; they offer high sensitivity but are subject to memory bias and low temporal resolution (Diarra et al., 2025).

2.2. Physiological Measures

Physiological measures assess MWL by analyzing individuals' physiological responses while they perform a primary task. Compared with subjective measures, physiological measures offer a more objective assessment because physiological signals directly reflect autonomic nervous system activity and are difficult to manipulate deliberately. Charles and Nixon (2019) systematically reviewed six categories of physiological measures: electrocardiac activity, respiration, skin-based measures, blood pressure, ocular measures, and brain activity.

The physiological basis of MWL measurement lies in the autonomic nervous system (ANS) response to cognitive demands. When an individual experiences increased MWL, sympathetic nervous system activation rises, producing measurable changes in cardiovascular, electrodermal, ocular, and neurophysiological signals (Charles & Nixon, 2019). Specifically, increased MWL triggers elevated heart rate, reduced heart rate variability, increased skin conductance, pupil dilation, changes in blink rate and fixation patterns, and altered brain oscillations, such as increased theta power and decreased alpha power (Tierney et al., 2025). These physiological changes form the foundation for inferring MWL from bodily signals.

(a) Electrocardiographic (ECG) Measures

ECG techniques measure the electrical activity of the heart. Heart rate (HR) and heart rate variability (HRV) are the most commonly used metrics derived from ECG data for assessing the effects of MWL on cardiovascular function. Heart rate is regulated by the balance between sympathetic nervous system (SNS) and parasympathetic nervous system (PNS) inputs to the sinoatrial node: SNS activation accelerates heart rate, whereas PNS activation slows it. Under increased MWL, SNS dominance leads to elevated heart rate and reduced HRV (Tierney et al., 2025).

HR has been shown to be sensitive to variations in MWL, with studies in simulated flight, computer operation, and simulated air traffic control demonstrating that HR increases with task demands, memory load, and task difficulty. HRV measures beat-to-beat variability and comprises time-domain and frequency-domain features. Commonly used time-domain features include the RR interval (i.e., the time between consecutive heartbeats), the percentage of successive intervals differing by more than 50 ms (pNN50), the standard deviation of normal-

to-normal (NN) intervals (SDNN), and the root mean square of successive differences (RMSSD), whereas the ratio of low- to high-frequency power (LF/HF) is widely used in the frequency domain. In general, HRV is inversely related to sympathetic nervous system arousal: as MWL increases, sympathetic activity and arousal rise, and HRV decreases. Specifically, increases in MWL are typically associated with decreases in RR, pNN50, SDNN, and RMSSD, and with an increase in LF/HF (Diarra et al., 2025). For instance, Wei et al. (2014) found that SDNN was sensitive to different levels of MWL in a simulated flight task, decreasing as MWL demands increased. Notably, findings regarding the direction of RMSSD are inconsistent across reviews: Tao et al. (2019) found that 78% of the surveyed studies reporting this measure showed a positive association with MWL.

ECG signals are often sampled at rates between 250 and 500 Hz. Artifacts such as muscle noise and electrode movement are removed, and abnormal beats are excluded from the analysis, particularly when computing NN intervals (Charles & Nixon, 2019). HRV is typically analyzed using statistical methods such as analysis of variance (ANOVA) to compare conditions (Backs, 1995; Brookings et al., 1996; Veltman & Gaillard, 1998). More advanced approaches, such as neural networks, have also been explored for classifying MWL states from cardiac features (Hogervorst et al., 2014; Wilson & Russell, 2003). Machine learning algorithms such as linear discriminant analysis (LDA), support vector machines (SVM), and k-nearest neighbors (KNN) have been used to classify MWL levels from ECG features (Mohanavelu et al., 2022). Findings are often evaluated by comparing physiological changes across different task-load conditions; for instance, HR increases with task demands (De Rivecourt et al., 2008), and a reduction in mid-frequency band power reflects an increase in MWL (Veltman & Gaillard, 1998). However, sensitivity can vary: HRV may distinguish between work and rest but not necessarily between adjacent levels of task demand (Nickel & Nachreiner, 2003).

(b) Respiratory Measures

Respiratory rate (BR), defined as the number of breaths per unit time, is the primary respiratory measure for MWL assessment. In general, BR increases with increasing MWL (Lean & Shan, 2012). Oxygen consumption has also been reported to exhibit a roughly positive linear relationship with MWL (Cárdenas-Vélez et al., 2013).

Respiratory signals are filtered to remove high-frequency noise. When speech is present, respiratory patterns can be altered, introducing artifacts unrelated to MWL (Bernardi et al., 2000). Physical activity should be kept as constant as possible, as changes in physical load also affect BR. Differences in respiratory rate between task conditions are typically examined using statistical comparisons. Research has shown that respiratory rate increases with task difficulty (Backs, 1995; Brookings et al., 1996; Fairclough & Venables, 2006). Results are often evaluated by correlating changes in respiratory rate with manipulated task load or subjective ratings. However, respiratory measures are highly dependent on physical activity and speech, which limits their use in certain applied settings (Charles & Nixon, 2019).

(c) Electrodermal Activity (EDA)

EDA is reflected in skin conductance and depends on sweat gland activity. It can be decomposed into two components: the slowly varying skin conductance level (SCL), considered the baseline, and the rapidly changing, event-related skin conductance response (SCR). As MWL increases, the sympathetic nervous system is activated, physiological arousal and sweat gland activity rise, and skin conductance increases. Kajiwaru (2014) found that SCL significantly reflected MWL levels under the influence of operation speed in a simulated driving study.

EDA signals are typically filtered to remove slow drift and high-frequency noise. Factors such as temperature, humidity, age, and sex can influence EDA, making comparisons across studies challenging (Kramer, 1990). There is a delay of 1-3 seconds between the eliciting stimulus and the observable skin conductance response. Statistical methods are used to compare EDA across different workload levels. Evidence suggests that EDA increases with task difficulty (Mehler et al., 2009) and is sensitive to sudden events, such as emergency braking (Collet et al., 2014). EDA has demonstrated good test-retest reliability in some multi-attribute tasks (Miyake et al., 2009). However, it may be insensitive to gradual changes in MWL and may exhibit a measurement ceiling

effect (Charles & Nixon, 2019). Although EDA is low-cost and provides effective biofeedback, its measurements are easily influenced by ambient temperature and humidity.

(d) Blood Pressure

Blood pressure is expressed as systolic and diastolic pressure. An increase in BP has been associated with increased task load (Charles & Nixon, 2019). However, BP is less frequently used than HR and HRV, which are less invasive and more sensitive to changes in MWL. BP measurements are also affected by many state variables, including physical activity, sleep, digestion, time of day, and speech (Adams & Leverland, 1985), and careful control of these factors is required during experimental design. BP is typically analyzed using repeated-measures ANOVA to compare different task phases or conditions. For example, BP has been found to be lowest during rest and highest during subjectively effortful tasks (Veltman & Gaillard, 1998). Although BP can differentiate between work and rest periods, it may not always increase with task complexity, as observed in a nuclear monitoring task (Hwang et al., 2008). Its sensitivity to MWL therefore appears to be context-dependent.

(e) Eye Tracking (ET) and Ocular Measures

Ocular measures include blink rate, blink closure rate, gaze angle, pupil size, pupil diameter, and pupillary responses. Indices such as blink rate, pupil diameter, fixation duration, and saccade duration have shown high sensitivity to changes in MWL (Diarra et al., 2025). For example, Gao et al. (2013) found that, in response to emergency situations in a digital nuclear power plant, participants' blink rate was sensitive to changes in task complexity, pupil diameter correlated strongly with error rate, and blink rate decreased while pupil diameter increased as task complexity increased. Marchitto et al. (2016) found that saccade duration was significantly shorter during an air traffic control conflict detection task with higher MWL. A key drawback of these measures is the difficulty of isolating the effects of visual workload from those of mental workload; some researchers argue that they are effective only for estimating visual workload.

Pupil diameter is affected by ambient illumination; therefore, changes in lighting conditions must be controlled or measured. Blink-related measures require accurate detection of blink events from electro-oculography (EOG) or video data (Charles & Nixon, 2019). Statistical comparisons are commonly used to examine differences in ocular measures across workload conditions. Blink rate decreases during visually demanding tasks (Brookings et al., 1996; Veltman & Gaillard, 1998; Wilson, 2002), and pupil diameter increases with task difficulty (Causse et al., 2010) and correlates with error rate (Gao et al., 2013). Ocular measures are considered good indicators of MWL, especially in visual tasks. However, they are susceptible to external factors such as light, air quality, and drugs. In addition, changes in blink rate may be observable only during the initial phase of a long-duration task (Fairclough et al., 2005).

(f) Electromyography (EMG)

EMG records the electrical potentials that drive skeletal muscle fiber contractions. EMG amplitude can help assess emotional states such as stress and tension, as well as MWL. For example, in a field study of traffic monitoring, Fallahi et al. (2016) found that EMG amplitude increased proportionally with MWL.

(g) Skin Temperature (ST)

Skin temperature measured at different body parts provides another method for assessing MWL, although it has not yet been widely applied. For example, Murai and Hayashi (2008) proposed nasal-forehead (N-F) temperature as an effective index for assessing navigators' MWL. Research has also shown that nasal tip temperature decreases when individuals perform secondary tasks (Itoh, 2009). Like EDA, ST measurements are influenced by changes in ambient temperature.

(h) Neuroendocrine Measures

Neuroendocrine measures, primarily salivary cortisol, have been used to reflect MWL levels. However, this approach is relatively rare because salivary cortisol responds only when the demands of the primary task increase substantially and shows little change under simple task conditions, giving it low sensitivity to graded

differences in MWL. More recently, this index has been applied mainly to measure stress (Charles & Nixon, 2019).

(i) Electroencephalography (EEG)

EEG records voltage fluctuations arising from ionic current differences within brain neurons via electrodes placed on the scalp. It is one of the most widely used methods for MWL assessment and is widely accepted because it directly measures signals from the brain rather than inferring MWL indirectly from other physiological responses. EEG signals are typically divided into five frequency bands: Delta (0–4 Hz), Theta (4–8 Hz), Alpha (8–13 Hz), Beta (13–39 Hz), and Gamma (>40 Hz). A common observation is that, as MWL increases, frontal Theta power increases and parietal Alpha power decreases. Frey et al. (2014) noted that the Alpha band is associated with attention, and its amplitude increases when participants are fatigued or have their eyes closed. Another indicator of MWL is the Theta/Alpha ratio, calculated from Theta power in frontal EEG channels and Alpha power in parietal EEG channels (Di Flumeri et al., 2018). Diaz-Piedra et al. (2020) found that, under the most complex task conditions, Theta power was higher in the frontal, temporal, and occipital regions of army drivers.

EEG can also capture event-related potentials (ERPs), which are electrophysiological responses to sensory, cognitive, and motor stimuli. The most frequently used ERP in the MWL literature is the P300, a positive deflection occurring approximately 300 ms after a stimulus. An inverse relationship has been established between MWL and P300 amplitude: as MWL increases, P300 amplitude decreases (Solís-Marcos & Kircher, 2019).

EEG signals are filtered to remove artifacts such as eye blinks, muscle activity, and line noise. Data are often epoched around events of interest for ERP analysis, and power spectral density is computed over specific frequency bands for frequency-domain analysis. Statistical methods are used to compare EEG power or ERP amplitudes across conditions. Alpha power suppression has been associated with increased cognitive demand (Brookings et al., 1996; Fournier et al., 1999). Classification approaches, such as neural networks, have achieved high accuracy in predicting individual states from EEG features (Hogervorst et al., 2014; Wilson & Russell, 2003). Machine learning algorithms such as LDA, SVM, and KNN have also been used to classify MWL levels from EEG features (Mohanavelu et al., 2022).

EEG measures have shown sensitivity to changes in MWL, with classification accuracies of up to 90% reported in some studies (Hogervorst et al., 2014). However, EEG measurement and analysis remain complex, the equipment is expensive, and participants' movements are relatively restricted. EEG is therefore more suitable for strictly controlled laboratory environments and for studies involving brain cognitive mechanisms (Diarra et al., 2025).

(j) Functional Near-Infrared Spectroscopy (fNIRS)

fNIRS is a non-invasive neuroimaging technique that uses low-level, non-ionizing light to record changes in brain activity. Through optical sensors placed on the scalp, fNIRS captures changes in cerebral blood flow, enabling researchers to study brain activation across various tasks. For example, Causse et al. (2017) had pilot cadets perform tasks of varying difficulty in a laboratory flight simulator while monitoring their prefrontal cortex activity with fNIRS. They found that as task difficulty increased, oxygenated hemoglobin (HbO₂) concentration in the prefrontal cortex increased, whereas deoxygenated hemoglobin (HHb) concentration decreased, indicating that fNIRS can effectively detect changes in brain activity under different MWL levels.

Compared with EEG, fNIRS offers higher spatial resolution, whereas EEG provides higher temporal resolution. Aghajani et al. (2017) compared the performance of fNIRS features alone with that of combined EEG–fNIRS features for MWL classification and suggested that, when developing passive brain–computer interfaces (BCIs) or other applications requiring monitoring of user MWL, combining EEG and fNIRS should be prioritized over either modality alone. Hamann and Carstengerdes (2022) investigated the effects of different task difficulties in a simulated flight environment and found that higher task difficulty led to higher subjective

MWL ratings, poorer task performance, increased frontal theta activity, and decreased frontal HHb. Both EEG and fNIRS could distinguish all induced MWL levels, with fNIRS being more sensitive to low-difficulty tasks and EEG more sensitive to high-difficulty tasks.

2.3. *Multimodal Measurement and MWL Assessment*

In recent years, the number of multimodal measurement studies—those employing multiple physiological measurement methods or combining task performance, subjective, and physiological measurements—has gradually increased. As Charles and Nixon (2019) pointed out, different measurement methods should be used complementarily, as each method answers only part of the question and reflects only part of an individual's mental workload. For example, in a study comparing the impact of Level 2 autonomous driving and manual driving on drivers' mental workload, Stapel et al. (2019) combined scale evaluation, secondary-task performance measurement, and physiological measurement to assess drivers' mental workload, and observed moderating effects of traffic flow complexity and drivers' autonomous driving experience. Specifically, the scale evaluation used the NASA Task Load Index (NASA-TLX); the secondary-task performance measurement employed an auditory detection response task, with reaction time and omission rate as indicators; and the physiological measurement included two indicators, SDNN and LF/HF.

Beyond using multimodal measurements to reflect an individual's mental workload from multiple perspectives, researchers have also begun to extract key features from different measurement data and combine them with machine learning or deep learning algorithms to construct predictive evaluation models of mental workload, thereby better characterizing the individual's current task execution and mental workload state.

(a) Multiple physiological measures combined with machine learning: Mohanavelu et al. (2022) used machine learning to fuse two types of physiological measures (EEG and ECG) to identify fighter pilots' MWL levels across different flight phases. They manipulated visibility and cognitive tasks in a high-fidelity flight simulator to simulate four MWL levels. Features extracted from EEG included power in the δ , θ , α , and β bands, band ratios, and non-linear features. Features extracted from ECG included time-domain features (RR interval, SDNN, RMSSD), frequency-domain features (LF power, HF power, LF/HF ratio), and non-linear Poincaré plot features (SD1, SD2, SD1/SD2). Principal component analysis (PCA) and the Friedman test were used to select significant features, and LDA, SVM, and KNN algorithms were employed. The results showed that, after feature selection, HRV features effectively distinguished MWL levels during the takeoff and landing phases, whereas EEG features effectively distinguished MWL levels during the cruise phase. LDA and SVM were more stable and accurate than KNN, and feature selection improved classification accuracy.

(b) Subjective + multiple physiological measures with machine learning: Lin and Lukodono (2022) used subjective and physiological measures to assess MWL in an industrial assembly scenario involving human–robot collaboration. The subjective measure was the NASA-TLX; physiological measures from wearable devices included HR, HRV, blood volume pulse (BVP), BVP amplitude, galvanic skin response (GSR), hand temperature, blood oxygen saturation (SpO₂), and blood pressure. KNN, SVM, Naïve Bayes, decision tree (DT), and random forest (RF) algorithms were used to classify MWL into three levels (low, optimal, and overload). The results showed that GSR, HR, BVP, and hand temperature were significantly correlated with MWL. RF achieved the best performance, with a classification accuracy as high as 94.73%.

Similarly, Lee et al. (2024) used subjective measures (Bedford Scale) and multiple physiological measures (ECG, EMG, EDA, EEG, and ET) to examine the effects of advanced driver assistance systems on driver MWL. Rather than extracting specific features, they used raw physiological data, preserving the natural shape and dynamics of the signals. A functional linear regression model was used to build a three-level MWL classification model. The results showed that ECG and EDA effectively predicted driver MWL, and 9 of the 31 signal combinations achieved prediction accuracies above 90%.

(c) Task performance + subjective + multiple physiological measures with machine learning: Singh et al. (2021) investigated which physiological features most effectively reflect MWL in a manned-unmanned teaming (MUM-T) context, combining self-reports using the Instantaneous Self-Assessment (ISA) scale, behavioural measures, and physiological measures. Seven machine learning algorithms were used: KNN, DT, AdaBoost, Gaussian Naïve Bayes (GNB), LDA, quadratic discriminant analysis (QDA), and SVM. Two validation methods were compared: a traditional method and an ecological method. The results showed that ECG features achieved the highest classification accuracy, followed by EEG and ET features. The study also demonstrated that the validation method significantly affected classification accuracy.

Moreover, Iqbal et al. (2024) recently demonstrated that fusing eye-tracking and EEG metrics improves multi-class classification of control room operators' cognitive workload, with fusion models outperforming single-modality approaches. Similarly, Yang et al. (2024) developed an attention-enabled learning framework with multimodal information fusion for real-time recognition of driver cognitive workload, achieving improved performance over single-modality baselines. Zhou et al. (2020) investigated multimodal physiological signals for workload prediction in robot-assisted surgery, finding that sensor fusion can enhance prediction accuracy in complex operational environments. Table 1 summarizes the signals, task contexts, models, reported performance, and validation protocols of the key empirical studies reviewed in this paper, together with validity-relevant observations for each study.

Table 1. Summary of representative studies on physiological signal-based mental workload (MWL) assessment cited in this paper.

Study	Signals	Task / Context	Model	Reported Performance	Validation Protocol	Validity-Relevant Note
Mohanavelu et al. (2022)	EEG + ECG	Fighter-pilot flight simulator; 4 MWL levels	LDA, SVM, KNN (PCA + Friedman feature selection)	LDA and SVM more stable and accurate than KNN (no figure given)	—	HRV features best during takeoff/landing; EEG features best during cruise
Lin & Lukodono (2022)	HR, HRV, BVP, GSR, hand temperature, SpO2, BP + NASA-TLX	Industrial assembly with human–robot collaboration; 3 MWL levels	KNN, SVM, Naïve Bayes, DT, RF	94.73% (RF, 3-class)	—	Labels derived from NASA-TLX; 120-min sessions; temporal alignment with ratings not analyzed
Lee et al. (2024)	ECG, EMG, EDA, EEG, ET (raw signals)	Driving with ADAS; 3 MWL levels	Functional linear regression	>90% for 9 of 31 signal combinations	—	Raw signals used without hand-crafted features; ECG and EDA most predictive
Singh et al. (2021)	ECG, EEG, ET + ISA self-report	MUM-T context	KNN, DT, AdaBoost, GNB, LDA, QDA, SVM	ECG features achieved the highest accuracy	Traditional (random) vs. ecological (time-based) splitting	Validation method significantly affected reported accuracy
Iqbal et al. (2024)	Eye-tracking + EEG	Control-room operators; multi-class workload	Fusion models	Fusion outperformed single-modality approaches	—	—

Yang et al. (2024)	Multimodal (fusion)	Real-time driver cognitive workload	Attention-enabled multimodal fusion	Improved over single-modality baselines	—	—
Ding et al. (2020)	Multimodal (EEG, ECG, etc.)	Simulated computer tasks (n-back); task difficulty as label	Random Forest	99.19% (binary)	Within-dataset	Only 10 participants; 3-s averaging window; goal stated as classifying n-back difficulty
Aksu et al. (2024)	EEG	n-back; 4-class	LightGBM	76.59% (4-class)	—	Most errors on 1-back trials, suggesting task-label proxy limitations
Grimes et al. (2008)	EEG	—	Naïve Bayes	99% (binary); 88% (4-class)	—	—
Subasi (2005)	EEG	Alertness-level recognition	ANN with wavelet transform	87–98%	—	—
Beh et al. (2021)	Wearable sensors (MAUS dataset)	n-back	—	—	5-fold CV vs. more realistic schemes	5-fold CV may overestimate performance
Sciaraffa et al. (2021)	EEG (passive BCI)	Mental-effort estimation	—	Intra-subject AUC 0.98; cross-subject AUC > 0.9 only at 45-s resolution; calibration-free neurometric AUC ≤ 0.73	Within-subject CV vs. LOSO	Cross-subject generalization bottleneck
Khan et al. (2024)	fNIRS + eye-tracking + vehicle dynamics	Driving simulator	CNN-LSTM	99.82%	—	Multimodal spatiotemporal fusion (arXiv preprint)
Zhang et al. (2024)	fNIRS	Pilot workload in flight simulation (takeoff and climb)	CNN + attention mechanism	88.69%	—	—
Ding et al. (2025)	Multimodal (DEAP dataset)	Emotion recognition	Multi-head cross-attention	94.88% valence; 95.26% arousal	—	Emotion recognition rather than MWL; attention weights modality contributions
Ng & Guan (2024)	EEG	Motor imagery (binary)	Few-shot subject-independent meta-	88.70%	Subject-independent	Rapid adaptation with minimal calibration

			learning			
Han et al. (2024)	EEG (META-EEG)	Motor imagery, zero-calibration	Gradient-based meta-learning with intermittent freezing	Outperformed baselines by 12%	Zero-calibration cross-subject	—
Manawadu et al. (2018)	—	Driver workload; multiclass	LSTM recurrent neural network	—	—	Temporal models capture sequential dependencies missed by static windows
Ayres et al. (2021)	Multiple (validity analysis)	Various task types	—	Blink rate 91.5% vs. mean pupil diameter 68.5%; detection rates 90% (memory/arithmetic) vs. 69% (simulation) and 67% (object/shape)	—	Sensitivity varies by measure and by task type
Johnson et al. (2014)	Psychophysiological metrics	Deadly-force judgment task; experts vs. novices	Regression	Up to 72% of variance explained (experts) vs. 37% (novices)	—	Expertise moderates the physiology–performance mapping

Note. All figures are as reported in the manuscript text; “—” indicates the item is not specified in the text. *LOSO* = leave-one-subject-out cross-validation; *CV* = cross-validation; *ADAS* = advanced driver assistance systems; *MUM-T* = manned–unmanned teaming; *BCI* = brain–computer interface.

2.4. Summary

Each class of MWL measurement methods has distinct advantages and limitations. Prasetyo and Iridiastadi (2025) noted that psychophysiological techniques currently serve as primary indicators in MWL assessment, although physiological measures are most frequently used in isolation, followed by self-reports and task performance measures. The most frequent combination pairs task performance measures with self-reports, and the joint application of all three classes of measures has proliferated over the past 15 years. Table 2 summarizes the typical direction of change as MWL increases, the sensitivity, and the key limitations of each physiological measure reviewed in Section 2.2.

Task performance measures provide direct, objective indicators of functional capacity but are limited by their inability to distinguish the source of workload variations and their susceptibility to non-workload factors. Primary task measures may fail to detect changes when task demands remain within an individual's capacity, whereas secondary task measures, although capable of quantifying spare capacity, are intrusive and detect only substantial workload changes. Self-report measures offer high sensitivity and diagnosticity with minimal intrusiveness, but their post-task administration introduces memory bias, low temporal resolution, and potential reliability issues in extended tasks. Physiological measures provide continuous, objective data that are difficult to deliberately manipulate, yet each modality carries specific constraints: cardiovascular measures are sensitive to physical activity, electrodermal measures are influenced by ambient conditions, ocular measures confound visual and mental workload, and neurophysiological measures require expensive equipment and controlled environments. Multimodal approaches, increasingly adopted in recent years, integrate these complementary

sources to mitigate individual limitations and enhance measurement validity. Nevertheless, combining multiple measures does not automatically resolve fundamental issues of construct validity, and the practical challenges of synchronizing diverse data streams should not be underestimated.

Section 2 has reviewed the conceptual foundations and current measurement approaches for MWL. Although MWL is widely recognized as a dynamic construct involving the interaction between task demands and available cognitive resources (Prasetyo & Iridiastadi, 2025), the validity and generalizability of physiological measurement remain uncertain. The following section examines the methodological pipeline from signal acquisition to model evaluation, identifying specific points at which validity may be compromised.

Table 2. Direction of change, sensitivity, and key limitations of physiological measures as mental workload (MWL) increases.

Physiological measure	Direction as MWL increases	Sensitivity and key limitations	Key supporting studies
Heart rate (HR)	↑	Sensitive to task demands, memory load, and difficulty in flight, computer, and ATC tasks; distinguishes work from rest better than adjacent demand levels	(De Rivecourt et al., 2008; Tierney et al., 2025; Veltman & Gaillard, 1998)
Heart rate variability (HRV)	RR, pNN50, SDNN, RMSSD ↓; LF/HF ↑	SDNN decreases with MWL (Wei et al., 2014); RMSSD direction inconsistent across reviews (78% positive association; Tao et al., 2019); may not separate adjacent demand levels	(Diarra et al., 2025; Tao et al., 2019; Wei et al., 2014)
Respiratory rate (BR)	↑ (O ₂ consumption rises ~linearly)	Strongly confounded by speech and physical activity	(Backs, 1995; Fairclough et al., 2005; Lean & Shan, 2012)
Electrodermal activity (EDA: SCL, SCR)	↑	Sensitive to sudden events (e.g., emergency braking); 1–3 s response delay; insensitive to gradual MWL changes; ceiling effect; affected by ambient temperature and humidity	(Collet et al., 2014; Kajiwara, 2014; Mehler et al., 2009)
Blood pressure (BP)	↑ with task load	Invasive; context-dependent sensitivity; may not increase with task complexity	(Charles & Nixon, 2019; Hwang et al., 2008; Veltman & Gaillard, 1998)
Pupil diameter	↑; correlates with error rate	Confounded by ambient illumination; visual workload and mental workload are difficult to isolate	(Causse et al., 2017; Gao et al., 2013)
Blink rate	↓ under visual demand (saccade duration also shortens)	Sensitive to task complexity; changes may appear only in the initial phase of long-duration tasks	(Fairclough et al., 2005; Gao et al., 2013; Marchitto et al., 2016)
Electromyography (EMG) amplitude	↑ proportionally with MWL	Also reflects emotional stress and tension (affective confound)	(Fallahi et al., 2016)
Skin temperature (ST)	Nasal-tip temperature ↓ during secondary tasks	Not yet widely applied; affected by ambient temperature	(Itoh, 2009; Murai & Hayashi, 2008)
EEG (θ , α , θ/α , P300)	Frontal θ ↑; parietal α ↓; θ/α ratio ↑; P300 amplitude ↓	Classification accuracy up to 90%; expensive equipment, restricted movement, suited to controlled laboratory settings	(Diaz-Piedra et al., 2020; Frey et al., 2014; Hogervorst et al., 2014)
fNIRS (HbO ₂ , HHb)	Prefrontal HbO ₂ ↑; HHb ↓	More sensitive at low difficulty (EEG more sensitive at high); lower temporal resolution than EEG; best used in combination with EEG	(Aghajani et al., 2017; Causse et al., 2017; Hamann &

Salivary cortisol	↑ only for substantial demand increases	Low sensitivity to graded MWL differences; now applied mainly to stress measurement	Carstengerdes, 2022) (Charles & Nixon, 2019)
-------------------	---	---	---

3. Methodological Challenges in Physiological MWL Assessment

The review in Section 2 indicates that, while current methods have advanced, the validity and generalizability of physiological MWL assessment remain uncertain: some studies report high classification accuracies, yet whether these models genuinely measure MWL requires closer examination. This section examines methodological issues at each stage of the physiological signal processing and machine learning pipeline—from raw signal acquisition to model evaluation—and proposes a reconceptualization of how MWL can be measured and identified using physiological signals and machine learning.

Although these stages are often treated as standard technical procedures, each involves methodological assumptions and design choices that can substantially influence the interpretation and validity of MWL measurement. We therefore also consider sources of variability that lie outside the technical pipeline itself. Specifically, inter-individual differences, intra-individual temporal dynamics, emotional states, and expertise can each threaten validity independently of the technical pipeline; these factors are examined in Section 3.7. Fig. 1 summarizes these challenges: the measurement pipeline and the formal limitation associated with each stage (left), and the sources of human variability that threaten validity beyond the pipeline (right).

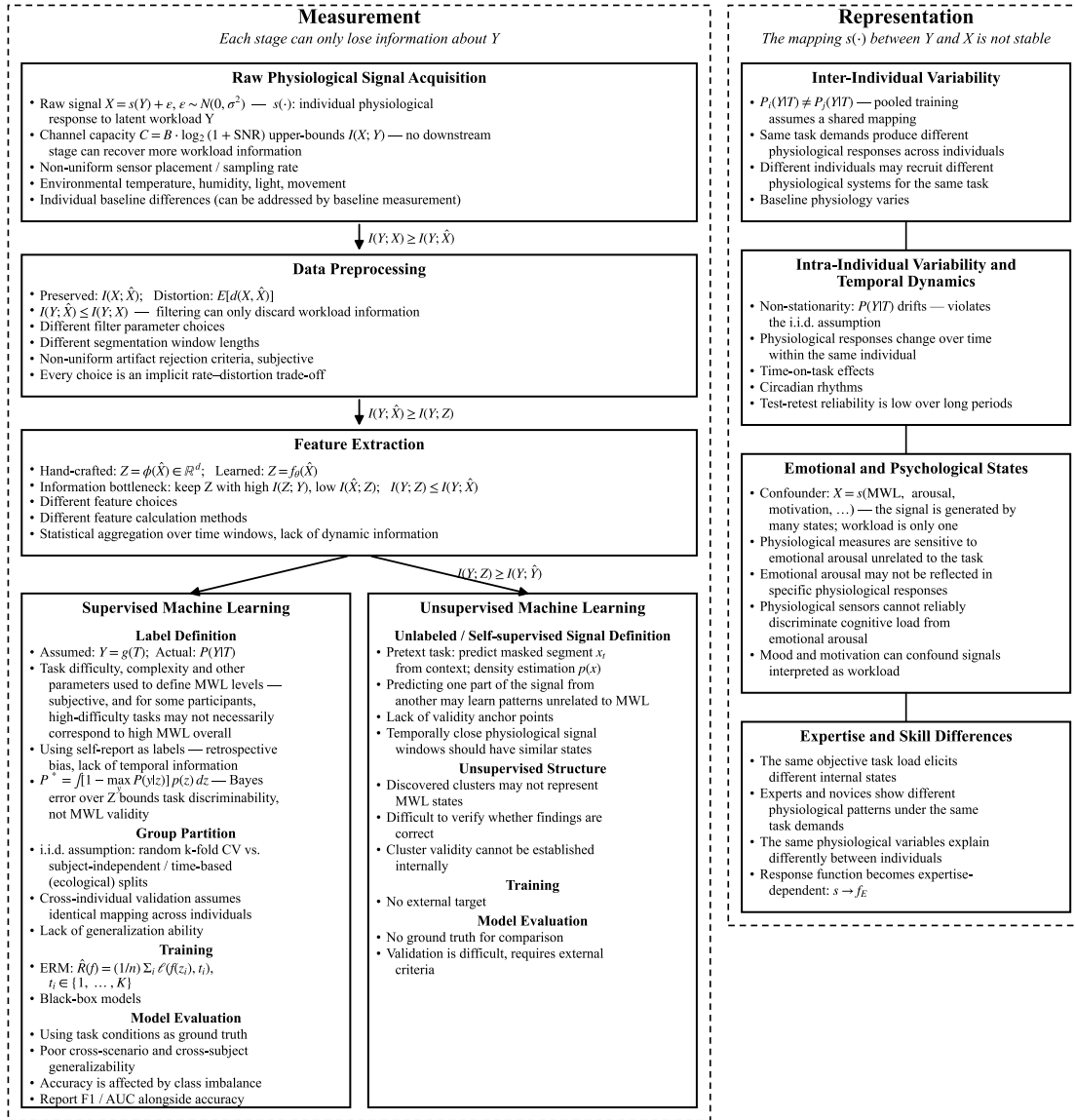


Fig. 1. Methodological and non-methodological challenges in physiological signal-based MWL assessment. Left: the measurement pipeline, with each stage annotated by its formal limitation. Right: representation challenges expressed as variability in $P(Y|T)$ across individuals, time, and psychological states. The chain $I(Y;X) \geq I(Y;\tilde{X}) \geq I(Y;Z) \geq I(Y;\hat{Y})$ (data-processing inequality) links the four stages: information about workload can only be lost, never created. Notation follows Section 3.

3.1. Stage 1: Raw Physiological Signal Acquisition

Sensor placement influences which physiological signals are captured and how they are interpreted, and configurations differ across studies. Cheema et al. (2018) used a 14-channel Emotiv Epoc+ headset with

electrodes placed according to the international 10–20 system, and Aksu et al. (2024) used a comparable 14-channel Emotiv device (EPOC X). Lin and Lukodono (2022) attached a Nexus-4 sensor to the upper arm and fingers for GSR and BVP and a smartwatch to the wrist for temperature and SpO₂. Variations in sensor placement—whether skin-conductance electrodes are placed on the hand, fingers, wrist, or foot—directly affect signal characteristics such as amplitude and signal-to-noise ratio, making cross-study comparison difficult. Scrivener and Reader (2022) further noted that EEG electrode positions vary across individuals, affecting signal characteristics and introducing systematic bias.

The choice of sampling rate determines which physiological phenomena are captured and which are missed. Cheema et al. (2018) and Aksu et al. (2024) used a sampling rate of 128 Hz for EEG, whereas Lin and Lukodono (2022) used 32 Hz for GSR and BVP and 1 Hz for temperature and SpO₂. A sampling rate that is too low may miss transient physiological events, whereas an excessively high rate introduces noise and computational burden. No consensus exists on optimal sampling rates for MWL assessment across physiological modalities (Fawwaz et al., 2024).

Environmental conditions influence physiological signals independently of MWL. Pupil dilation, for example, is sensitive to changes in ambient light. Ayres et al. (2021) noted that researchers studying pupil dilation must maintain constant stimulus luminance and use baseline data or computational correction methods; otherwise, cognitive effects cannot be separated from light reflexes. Aksu et al. (2024) conducted their experiments in a sound-insulated, fluorescent-lit laboratory, yet even under such controlled conditions, environmental factors such as temperature and humidity can still influence EDA measurements (Charles & Nixon, 2019; Kramer, 1990).

Individual differences in baseline physiology mean that the same absolute physiological value may correspond to different MWL levels across individuals: a heart rate of 90 bpm may indicate high MWL for one individual but baseline for another. Ayres et al. (2021) noted that within-subjects designs help control for prior knowledge and individual differences. However, several machine learning studies—including Aksu et al. (2024) and Lin & Lukodono (2022)—train models on data pooled from multiple participants, thereby assuming a common mapping from physiology to MWL.

From an information-theoretic perspective, the signal acquisition process can be modeled as a noisy communication channel in which the true physiological state Y is transmitted through a measurement apparatus that introduces additive noise ϵ . The observed signal X is therefore a corrupted version of the true state, $X = s(Y) + \epsilon$, where $s(\cdot)$ denotes the individual physiological response function and $\epsilon \sim N(0, \sigma^2)$ captures sensor noise, environmental interference, and movement artifacts. The Shannon–Hartley theorem establishes a fundamental upper bound on the rate at which information about Y can be recovered from X : $C = B \log_2(1 + P_{\text{signal}}/P_{\text{noise}})$, where B is the signal bandwidth and $SNR = P_{\text{signal}}/P_{\text{noise}}$ is the signal-to-noise ratio (Shannon, 1948). Here, C denotes the maximum rate at which information about the physiological state can be transmitted through the measurement channel; it is a property of the acquisition hardware and environment, not of the classifier. When electrode impedance is high, ambient electrical noise is present, or movement introduces artifacts, P_{noise} increases and the channel capacity C decreases. Information about mental workload that is present in the original physiological process may thus be irrecoverably lost at the acquisition stage—no amount of downstream sophistication can recover what was never captured.

Consequently, signals acquired at this stage reflect a mixture of MWL-related activity, individual baseline differences, environmental influences, and measurement artifacts, and all subsequent analysis inherits these confounds. The bias at this stage is systematic rather than random: limited channel capacity attenuates workload-relevant variance relative to noise, so workload effects are more likely to be underestimated than overestimated. Reporting electrode impedance, signal-to-noise ratio, and standardized sensor placement (Scrivener & Reader, 2022), together with a baseline recording, provides a practical way to bound how much information could have been captured.

3.2. Stage 2: Data Preprocessing

These confounded signals then enter the preprocessing stage, where further methodological choices shape the data ultimately used for feature extraction and model training.

The choice of filter parameters determines which frequency components of the signal are preserved or removed. Aksu et al. (2024) applied a 1–50 Hz band-pass finite impulse response (FIR) filter followed by artifact rejection based on independent component analysis (ICA) using the ICLabel function, removing components with a probability greater than 90% of being ocular, muscular, or other artifacts. Lin and Lukodono (2022) used BioTrace+ software for data processing. A filter that is too aggressive may remove signal components relevant to MWL, whereas one that is too weak may retain noise. Ayres et al. (2021) noted that HRV spectral analysis requires specific filtering to separate respiratory effects (high frequency, 0.15–0.40 Hz) from blood pressure regulation (mid-frequency, 0.07–0.14 Hz). Filter parameters thus differ across studies, and no standardized protocol exists.

The length of the segmentation window determines whether rapid fluctuations in MWL are preserved or averaged out. Cheema et al. (2018) segmented the EEG data into 3-second epochs aligned with each stimulus presentation, Aksu et al. (2024) used stimulus-based segmentation with 2500 ms response windows, and Lin and Lukodono (2022) collected data continuously over 120 minutes without specifying segmentation parameters. Segmentation assumes that MWL remains stable within each window; however, as Longo et al. (2022) note in their definition, MWL “varies dynamically over time” and can change within seconds. A 2-minute window, as used in some studies, would average together periods of high and low MWL, smoothing out transient overload episodes that may be critical for detecting safety-critical states. Ayres et al. (2021) similarly noted that HRV measures are less effective for longer-duration tasks, because such tasks naturally contain both mentally active and inactive periods, producing a noisier signal.

The criteria for rejecting artifacts introduce subjectivity into data processing. Cheema et al. (2018) used FORCe, an automated method, but acknowledged that most artifact-removal methods either require additional EOG/EMG recordings or are designed to remove only single artifacts. For ECG signals, artifacts such as muscle noise or electrode movement are typically removed, and abnormal beats are excluded from the analysis, particularly when computing NN intervals (Charles & Nixon, 2019). The threshold for excluding a component—for example, the 90% probability threshold used by Aksu et al. (2024)—is ultimately a researcher's choice that can affect the results.

Preprocessing is not a neutral step: every filter, window, and artifact-rejection criterion makes an implicit decision about which aspects of the physiological signal are worth keeping. Rate–distortion theory formalizes the cost of these decisions as an unavoidable trade-off between the information preserved and the distortion introduced. Let X denote the raw acquired signal and $\hat{X}$ the preprocessed output after filtering, segmentation, and artifact rejection. The mutual information $I(X; \hat{X})$ quantifies how much information about the original signal is preserved, whereas the expected distortion $E[d(X, \hat{X})]$ measures the degree of alteration introduced by preprocessing (Cover & Thomas, 2006). For mental workload assessment, $I(X; \hat{X})$ quantifies how much MWL-relevant information survives preprocessing: when $I(X; \hat{X})$ is low, the preprocessed signal may no longer contain enough information to distinguish high from low workload. Conversely, $E[d(X, \hat{X})]$ measures how much the signal has been altered: high distortion means that transient workload peaks may have been smoothed out or individual baseline differences distorted.

Band-pass filtering with cutoff frequencies $[f_{low}, f_{high}]$ discards all spectral components outside this band, introducing a reconstruction error $\|X - \hat{X}\|_2^2 = \sum_{\{f \notin [f_{low}, f_{high}]\}} |X(f)|^2$. Similarly, segmentation into fixed-width windows of duration T effectively applies a temporal low-pass filter, attenuating workload fluctuations faster than $1/T$ Hz. The choice of filter parameters and window length thus determines which frequency components of the workload signal are preserved and which are discarded, with direct consequences for the

validity of subsequent inference. In practical terms, a 3-second averaging window (Cheema et al., 2018) does not merely ‘clean’ the data—it actively decides that workload fluctuations faster than 1/3 Hz are irrelevant. If an overload event lasts only one second, it may be irrecoverably lost at this stage, regardless of the accuracy of the downstream classifier.

The cumulative effect is that preprocessed signals represent a filtered, segmented, and artifact-conditioned version of the original physiological data, in which some high-frequency MWL fluctuations may have been systematically attenuated. This bias has a clear direction: smoothing and aggressive filtering preferentially remove transient peaks, so short-lived overload episodes are more likely to be missed than preserved. A sensitivity analysis—repeating the pipeline under alternative filter cutoffs and window lengths and checking whether the conclusions survive—is a feasible way to assess how large this bias is in a given study.

3.3. Stage 3: Feature Extraction

The preprocessed signals—now filtered, segmented, and artifact-conditioned—are then subjected to feature extraction, where decisions about which signal characteristics to retain and which to discard determine which aspects of the physiological response will be available for modeling.

The same physiological pattern may arise from different psychological states, and the same MWL state may be realized through different physiological configurations. Charles and Nixon (2019) noted that elevated heart rate and increased skin conductance can result from cognitive load, emotional stress, physical exertion, or a combination thereof. Tao et al. (2019) reported that, even when findings are aggregated across studies, 28% of physiological measure instances show no significant association with MWL. Together, these observations indicate that the relationship between physiological signals and MWL is not one-to-one but many-to-many.

In Lin and Lukodono (2022), GSR correlated strongly with MWL ($r = 0.83$), and HR, BVP, and hand temperature also showed significant correlations. However, correlation does not imply causation: Ding et al. (2020) reported no significant main effect of task difficulty on heart rate ($p = 0.086$), whereas skin conductance and respiration rate did show such effects. Thus, even for the same participants performing the same tasks, some physiological systems responded to MWL while others did not. Consequently, a classifier trained on one set of physiological features may fail to generalize to another feature set or to new individuals.

The choice of which features to extract determines which aspects of the physiological signal are retained for analysis. Cheema et al. (2018) extracted a comprehensive feature set spanning six categories (statistical, derivative, interval, Hjorth, frequency-domain, wavelet) and then applied minimum-redundancy maximum-relevance (mRMR) feature selection (Peng et al., 2005) to reduce the set to 15 optimized features. Aksu et al. (2024) extracted 70 EEG band power features plus 14 theta/alpha ratios, along with 27 eye-tracking features. Lin and Lukodono (2022) extracted 12 features including GSR, HR, BVP, HRV parameters, blood pressure, SpO₂, and hand temperature. Different studies therefore select different feature sets, with limited theoretical justification for why specific features should be sensitive to MWL and no consensus on which features are optimal. Roy et al. (2019), in their systematic review of deep learning-based EEG analysis, found that end-to-end learning approaches—which bypass hand-crafted feature extraction—can achieve competitive performance but still face generalizability challenges.

Extracting summary statistics discards temporal structure. Charles and Nixon (2019) noted that EDA is sensitive to sudden changes but not to gradual increases in MWL. A mean GSR value over 3 seconds cannot distinguish between a sudden peak and a gradual ramp. Temporal dynamics such as rise time, recovery rate, and frequency of fluctuations may carry information about MWL that mean values do not capture. Ayres et al. (2021) found that blink rate—a measure that retains temporal information about event frequency—was among the more sensitive physiological indicators (91.5% accuracy in their sample), whereas average-based measures such as mean pupil diameter showed lower sensitivity (68.5%). Lee et al. (2024) took a different approach,

using raw physiological data without extracting specific features, thereby preserving the natural shape and dynamics of the signals, and achieved prediction accuracies above 90% for some signal combinations.

From an information-theoretic perspective, feature extraction can be formalized as learning a representation mapping $f: X \rightarrow Z$ that compresses high-dimensional raw signals into a lower-dimensional feature space. The central question is which aspects of the physiological signal should be retained and which can be safely discarded. Retaining too little risks losing workload-relevant information; retaining too much risks introducing noise that confounds the classifier. The Information Bottleneck principle provides a theoretical framework for evaluating such representations: an optimal representation $Z = f(X)$ should minimize $I(X; Z) - \beta I(Z; Y)$, compressing irrelevant variability in X while preserving information about the target workload state Y (Tishby & Zaslavsky, 2015), where $I(X; Z)$ measures how much the original signal is compressed into the feature space (smaller values indicate stronger compression), $I(Z; Y)$ measures how informative those features are about the target workload state Y (larger values indicate better prediction), and β controls the balance between the two. Hand-crafted features embody explicit assumptions about which aspects of the physiological signal are workload-relevant; when these assumptions are incorrect, the representation becomes overly restrictive, either discarding task-relevant information (retaining too little) or preserving confounding noise (retaining too much). The shift toward end-to-end deep learning (Kingphai & Moshfeghi, 2025) can be understood as relaxing this manual bottleneck, allowing the model to learn the compression trade-off directly from data. However, without adequate regularization or sufficient training data, learned representations may overfit to dataset-specific artifacts rather than extracting generalizable workload signatures.

In sum, feature extraction involves choices with both theoretical and practical consequences. Different feature sets capture different aspects of the physiological response. As Kingphai and Moshfeghi (2025) noted in their systematic review of deep learning models for MWL assessment from EEG, the shift toward end-to-end learning aims to reduce information loss from hand-crafted features, but it introduces new challenges in interpretability and model validation. The bias here is selective blindness: mean-based features are insensitive to gradual changes, whereas event-based features miss slow drifts (Charles & Nixon, 2019). Comparing candidate measures before modeling—as in Ayres et al. (2021), where blink rate (91.5%) proved far more sensitive than mean pupil diameter (68.5%)—quantifies this bias empirically, and the sensitivity and diagnosticity criteria of Nickel and Nachreiner (2003) provide the classical framework for such comparisons.

3.4. Stage 4: The Labeling Problem in Supervised Learning

These extracted features are then paired with labels in supervised learning frameworks. The validity of the resulting model depends critically on whether the labels accurately reflect the construct of MWL—a question that has received insufficient attention in the literature.

Using task difficulty as a label assumes that task demand directly determines MWL. Ding et al. (2020) and Aksu et al. (2024) both used task difficulty as the label for supervised classification (arithmetic difficulty levels in Ding et al.; n-back levels in Aksu et al.). However, as Longo et al. (2022) define, MWL is mediated by external environmental and situational factors and is influenced by internal characteristics of the individual, such as skills and past experience. Task demand is only one such factor: the same task difficulty can produce different MWL levels in different individuals, or even in the same individual at different times.

Ayres et al. (2021) noted that intrinsic cognitive load depends on the learner's expertise; a task that is complex for a novice may be simple for an expert. Aksu et al. (2024) collected NASA-TLX scores but used them only for validation and correlation analysis, not as labels. In Ding et al. (2020), NASA-TLX scores in the difficult condition averaged 62.79 with a standard deviation of 14.75. Two participants in the same "difficult" condition could thus differ by 30 points on the NASA-TLX scale. A classifier trained to predict "difficult" from

physiological features would treat these two participants as identical, even though their experienced MWL differed substantially.

Using self-report ratings as labels introduces retrospective bias and temporal mismatch. Lin and Lukodono (2022) used NASA-TLX scores to categorize MWL into three classes (underload, optimal load, overload). Ayres et al. (2021) found that subjective measures had high validity compared with other approaches, with single-item measures achieving 100% accuracy in matching task performance in their sample. However, self-report ratings are typically administered post-task and may introduce memory bias or be influenced by the individual's perception of their own performance (Prasetyo & Iridiastadi, 2025). A participant who experienced high MWL only during the final minutes of a 120-minute task might report a high NASA-TLX score, yet the physiological signals averaged across the entire task would not show a clear MWL signature. Conversely, a participant who experienced moderate MWL throughout might report the same score. Lin and Lukodono (2022) collected data over 120 minutes but did not analyze the temporal alignment between physiological changes and NASA-TLX ratings.

Using task difficulty as a supervised learning label assumes a deterministic mapping $Y = g(T)$ between task demand T and mental workload Y . However, this mapping is fundamentally stochastic and context-dependent: the conditional distribution $P(Y | T)$ varies across individuals, situations, and time. From a statistical learning perspective, training a classifier on task-condition labels minimizes the empirical risk $\hat{R}(f) = 1/n \sum_{i=1}^n \ell(f(x_i), t_i)$, where $t_i \in \{1, \dots, K\}$ indexes the task condition. Yet even a Bayes-optimal classifier—one that makes the best possible predictions given the data—incurs an irreducible error bounded below by $P^* = \int [1 - \max_t P(t | x)]P(x)dx$, which reflects the discriminability of task conditions rather than the validity of workload inference (Cover & Thomas, 2006). This bound P^* has an important interpretation: it reflects how separable the task conditions are in physiological feature space, not whether those conditions correspond to different experienced workload levels. A model may achieve near-perfect task classification while capturing no meaningful information about the experienced cognitive state. In other words, P^* reveals that high classification accuracy on task-condition labels is a necessary but not sufficient condition for valid workload measurement.

Furthermore, even when labels are reasonably valid, the choice to treat workload as a discrete category introduces a second source of error. If the true workload $Y \in \mathbb{R}$ follows a continuous distribution, discretization into K classes with thresholds $\{c_1, \dots, c_{K-1}\}$ introduces quantization error $E[(Y - \hat{Y})^2]$, where $\hat{Y} = c_k$ for $c_{k-1} < Y \leq c_k$. This error is largest near class boundaries—the very regions where accurate workload assessment matters most for safety. For example, a model classifying "medium" versus "high" workload may assign the same "high" label to a workload level just 1% above the threshold and to one deep in the danger zone, even though these two situations require radically different interventions. By imposing discrete boundaries on a continuous phenomenon, the classifier discards relevant information about the magnitude of cognitive demand.

The true MWL state at each moment is not directly observable. This is a construct validity problem, not a data quality problem. As Borsboom et al. (2004) argued, validity requires that "variations in the attribute causally produce variation in the measurement outcomes." If the attribute (MWL) is not directly measurable, then supervised learning lacks a verifiable ground truth.

Task difficulty is an external manipulation, not a direct measure of experienced MWL. Self-report captures a retrospective summary rather than continuous dynamics. Unsupervised methods avoid the labeling problem but face their own validation challenges. Kumar Das et al. (2025) systematically reviewed ground-truth labeling practices for cognitive workload adaptive systems and concluded that the field lacks consensus on what constitutes valid ground truth, with task difficulty, subjective ratings, and performance metrics all serving as imperfect proxies. The bias introduced by proxy labels inflates apparent validity, because the model is rewarded for discriminating task conditions rather than workload states, and retrospective self-reports shift labels toward salient end-of-task moments. Concurrent validation against independent criteria and systematic audits of ground-truth labeling practices (Kumar Das et al., 2025) are the available assessment approaches.

3.5. *The Problem of Model Transferability*

Even when labels are reasonably valid, the trained model must generalize beyond the specific experimental context in which it was developed. The transferability of MWL classifiers across tasks, individuals, and domains represents a separate and equally important challenge.

Models that perform well within a specific experimental context typically fail to transfer to new contexts. Charles and Nixon (2019) emphasized that physiological responses to MWL are highly scenario-dependent. A classifier trained on simulated flight tasks may not work for driving or nuclear power plant operations (Veltman & Gaillard, 1998).

This transferability problem has two sources. First, the mapping from physiology to MWL differs across task domains. Tao et al. (2019) reported that the validity of specific measures varies by research domain. For example, cardiovascular measures were more effective in driving studies than in other domains, whereas eye movement measures were less effective in nuclear power studies. A classifier optimized for one domain captures domain-specific regularities that do not generalize.

Second, the mapping differs across individuals. Baseline physiological parameters vary substantially from person to person (see Section 3.1), yet most studies—including Ding et al. (2020) and Moustafa et al. (2017)—train models on data pooled from all participants, implicitly assuming a common mapping. Melnik et al. (2017) showed why this assumption is unlikely to hold: individual differences in physiological responses to MWL are substantial. A model fit to the group average may therefore systematically fail on individuals whose physiology deviates from that average, directly undermining cross-individual transfer.

Recent studies have provided concrete evidence of this transferability problem. Zhou et al. (2023) found that deep domain adaptation is necessary for EEG-based cross-subject cognitive workload recognition, implying that, without adaptation, models fail to generalize. Guan et al. (2023) demonstrated that cross-task MWL recognition based on EEG tensor representations requires transfer learning to achieve acceptable performance. Sciaraffa et al. (2021) reported that within-subject calibration reached an area under the curve (AUC) of 0.98 in mental effort estimation with passive BCIs, whereas leave-one-subject-out models had to lower the temporal resolution to 45 seconds to maintain an AUC above 0.9, and a calibration-free neurometric performed significantly worse, with the AUC not exceeding 0.73. These findings underscore that cross-subject generalization remains a major bottleneck in physiological MWL assessment. This transferability failure is not uniform across individuals: models err most on those whose physiological response patterns deviate from the pooled average, so the bias concentrates on atypical responders. Contrasting within-subject and leave-one-subject-out validation—as in the drop from within-subject calibration to the calibration-free neurometric reported by Sciaraffa et al. (2021)—directly quantifies this bias, and domain-discrepancy measures (Ben-David et al., 2010) provide a formal estimate.

3.6. *Stage 5: Model Training and Evaluation*

Finally, model performance must be evaluated using appropriate validation methods. The choice of evaluation protocol—whether random cross-validation, leave-one-subject-out, or time-based splitting—can substantially affect reported accuracy and, more importantly, the inferences drawn about the model's validity as a measure of MWL.

High classification accuracy on a specific dataset does not guarantee that the model captures MWL. Cheema et al. (2018) reported 99.19% accuracy for two-class classification using Random Forest. Aksu et al. (2024) reported 76.59% for four-class classification using LightGBM. Lin and Lukodono (2022) reported 94.73% using Random Forest. Grimes et al. (2008) reported 99% accuracy for two-class and 88% for four-class

classification using Naïve Bayes on EEG data. Subasi (2005) reported 87–98% using artificial neural networks (ANN).

However, a model that achieves high accuracy on task-condition classification may have learned to discriminate task difficulty levels rather than experienced MWL. Aksu et al. (2024) noted that their model made the most errors when classifying observations related to the 1-back task. One possible explanation is that the experienced MWL for the 1-back task varies more across individuals than for the 0-back or 3-back tasks. If so, the task-condition label is a poor proxy for actual MWL, and high accuracy on other class pairs may reflect the separation of task conditions that differ more clearly in demand rather than more valid measurement of MWL.

As noted in Section 3.5, training models on data pooled from multiple participants assumes a common mapping from physiology to MWL across individuals. Cheema et al. (2018) trained models on pooled data from 10 participants and tested them on held-out data from the same participants—a within-dataset, within-task evaluation. They did not test cross-participant generalization without retraining. Aksu et al. (2024) used 5-fold cross-validation within their 15-participant sample. Ayres et al. (2021) found that within-subject designs yield more valid results because they control for individual differences in prior knowledge and baseline physiology. A model trained on pooled data may therefore capture between-subject variance unrelated to MWL.

Models trained on one type of task may not generalize to others. Ayres et al. (2021) found that memory and arithmetic tasks produced higher detection rates for physiological measures (90% in their sample), whereas simulations and object/shape manipulations yielded lower rates (69% and 67%, respectively). Charles and Nixon (2019) noted that physiological responses to MWL are scenario-dependent. Ding et al. (2020) and Aksu et al. (2024) used, respectively, mental arithmetic tasks and n-back tasks, both controlled laboratory working-memory paradigms. Whether a classifier trained on data from such paradigms would generalize to driving, air traffic control, or other real-world tasks remains an open question. Zhao et al. (2018) attempted real-time cross-task MWL assessment using physiological measures during anomaly detection and found that cross-task generalization requires careful consideration of task-specific features.

The choice of validation method affects reported accuracy. Singh et al. (2021) compared a traditional method (randomly splitting data without regard to time) with an ecological method (splitting based on time, more closely resembling real-world applications). The validation method influenced the reported performance of the models. This suggests that reported accuracy figures may vary with the validation procedure. Beh et al. (2021) introduced the MAUS dataset for MWL assessment and found that 5-fold cross-validation may overestimate performance relative to more realistic evaluation schemes.

Using task conditions as the evaluation standard creates circular reasoning. If labels are derived from task difficulty, high classification accuracy confirms only that physiological signals vary with task difficulty—not that they measure MWL. Ding et al. (2020) stated their goal as "to classify the mental workload into different levels" based on arithmetic task difficulty. Aksu et al. (2024) similarly used task difficulty as the target variable. This circularity means that reported accuracy figures do not provide direct evidence of MWL measurement validity. As Zhou et al. (2022) argued in their comprehensive review, the field must move beyond reporting accuracy alone and establish construct validity as the primary evaluation criterion.

Treating MWL as a discrete category imposes arbitrary boundaries on a continuous phenomenon. Aksu et al. (2024) used 4 classes (0-back, 1-back, 2-back, 3-back) and also reported 3-class and 2-class results. Lin and Lukodono (2022) used 3 classes (underload, optimal load, overload) based on NASA-TLX cutoffs.

As Prasetyo and Iridiastadi (2025) emphasized, MWL is a dynamic construct that varies continuously over time in response to task demands and environmental factors. Cognitive resources are not allocated in discrete chunks. Discretization requires setting cutoffs. In Lin and Lukodono (2022), optimal load was defined as NASA-TLX scores between 50 and 70. A score of 49 would be classified as underload, whereas a score of 50 would be optimal load. Whether this one-point difference represents a meaningful difference in MWL is unclear. Different studies use different cutoffs, so a model trained on one set of boundaries may not generalize to another.

From a statistical learning perspective, when a model trained on one group of participants is applied to another, or when a model trained on simulated flight is applied to driving, the fundamental assumption of standard machine learning—that training and test data are drawn from the same distribution—is violated. Cross-subject and cross-task generalization failure can be formalized as a domain adaptation problem in which the training distribution $P_{S(x,y)}$ differs from the test distribution $P_{T(x,y)}$. Ben-David et al. (2010) established that target risk is bounded by the sum of three terms: training error $R_{S(f)}$, domain discrepancy $d_{H\Delta H}(P_S, P_T)$, and optimal joint error λ^* . Here, the domain discrepancy $d_{H\Delta H}(P_S, P_T)$ measures how different the training and test distributions are in the feature space: when physiological response patterns differ substantially across individuals or tasks—as demonstrated by Matthews et al. (2015) and Tao et al. (2019)—this term increases, and the bound becomes vacuous unless explicit domain adaptation is performed.

Domain adaptation addresses distribution mismatch, but a second problem arises even within a single domain: the relationship between model complexity and available data. For a hypothesis class H with VC dimension d_{VC} trained on n samples, the generalization error satisfies (Vapnik, 2013): $R(f) \leq \hat{R}_n(f) + O(\sqrt{(d_{VC} \log(n/d_{VC}))/n}))$. This bound reveals a fundamental constraint: as model complexity d_{VC} increases relative to sample size n , the gap between training accuracy and true generalization accuracy widens. Contemporary deep neural networks used for physiological signal classification frequently contain millions to billions of parameters, with Vapnik–Chervonenkis (VC) dimensions that scale with or exceed the number of parameters (Zhang et al., 2017). In contrast, physiological workload studies typically collect data from tens of participants performing hundreds to thousands of trials—for instance, the STEW mental workload dataset contains recordings from only 48 participants. This mismatch between model capacity and dataset size yields large generalization gaps, explaining why models that achieve high accuracy in within-dataset validation frequently fail on new subjects or tasks.

Discrete classification discards information about the magnitude of MWL. A correct classification of "high" MWL provides no information about whether the MWL level was just above the threshold or far into the overload zone. For safety-critical applications—monitoring pilots, air traffic controllers, or nuclear plant operators—the distinction between moderate and severe overload may be more important than that between medium and high. Regression approaches that predict NASA-TLX scores directly would retain this magnitude information. Evaluation-related bias runs upward: random k -fold cross-validation overestimates deployable accuracy relative to time-based or subject-wise splits (Beh et al., 2021; Singh et al., 2021). Reporting F1 scores and AUC alongside accuracy, and validating under ecological (time-based) splits, are the standard correctives.

3.7. Inter- and Intra-Individual Variability: Threats to Validity Beyond the Pipeline

The methodological challenges discussed in Sections 3.1 through 3.6 concern the technical pipeline of physiological MWL assessment, from signal acquisition to model evaluation. Even when this pipeline is technically sound, substantial threats to validity and reliability persist, arising from sources outside the measurement apparatus itself: the inherent variability of human physiology and psychology across individuals and within the same individual over time. Individual differences in baseline physiology, circadian rhythms, emotional states, and skill levels all modulate physiological responses to task demands in ways that cannot be controlled through improved sensors or algorithms alone. This section examines four such sources of variability that complicate the interpretation of physiological MWL data.

(a) Inter-Individual Variability

Even under identical experimental conditions, different individuals exhibit markedly different physiological responses to the same task demands. Matthews et al. (2015) demonstrated that multiple psychophysiological measures of workload, although individually sensitive to task demands, are psychometrically divergent and fail to converge on a general workload factor. In their study, some participants showed greater changes in brain

activity (EEG), whereas others showed more pronounced cardiac responses (ECG), indicating that the same task may engage different physiological systems in different individuals. This dissociation poses a fundamental challenge to cross-subject generalization, as a classifier trained on one individual's response pattern may fail on another. In a systematic review, Tao et al. (2019) further confirmed that although most physiological measures can discriminate changes in MWL, they are not universally valid across all task scenarios and populations. Marchand et al. (2021) noted that inter-subject variability—and sometimes intra-subject variability when multiple measurements are taken within a day—hampers the validity of conclusions drawn from physiological data. The physiological baseline itself varies with age, gender, cardiovascular fitness, and physical health, all of which modulate measures such as HRV and EEG alpha power independently of task demands. In machine learning terms, this variability manifests as a domain shift in which $P_i(x) \neq P_j(x)$ across subjects, violating the i.i.d. assumption underlying standard supervised learning and necessitating domain adaptation techniques to align divergent physiological distributions (Gretton et al., 2012). The resulting bias is systematic for atypical responders: individuals whose dominant physiological channel differs from the group pattern are consistently misclassified rather than randomly misclassified. Stratified reporting—per individual or per subgroup rather than pooled averages—makes this bias visible (Matthews et al., 2015).

(b) Intra-Individual Variability and Temporal Dynamics

The physiological response to a given task is not stable even within the same individual. Hidalgo-Muñoz et al. (2018) found significant time-on-task effects on cardiovascular measures, with heart rate decreasing and heart rate variability increasing over the course of a flight simulation owing to training and habituation effects. This means that the physiological signature of a high-workload condition may change within a single session, confounding attempts to map physiological states onto fixed workload levels. Feltman (2016) examined circadian effects on MWL and found that subjective workload measures were influenced by time of day, with a moderating effect of circadian typology. Although performance differences were not significant, differences in EEG and subjective workload emerged as a function of testing phase and workload manipulations. Getzmann et al. (2021) conducted a test-retest analysis of EEG-based workload assessment over 12–15 months and found that although group-level patterns were reliable, intra-individual correlations were low, with some participants even showing reversed patterns between sessions. Miyake et al. (2009) examined the test-retest consistency of physiological responses to mental tasks across three days and found that although some parameters showed significant correlations, large individual differences persisted. Together, these findings underscore that physiological MWL measures reflect not only task demands but also transient states such as alertness, vigilance, and familiarity.

These intra-individual changes are not merely random noise; they reflect systematic shifts in how the body responds to cognitive demands over time. Temporally, this constitutes concept drift, in which the mapping from physiology to workload evolves as $P_t(y | x) \neq P_{t+\Delta}(y | x)$, requiring online adaptation mechanisms that track the individual's changing physiological baseline rather than assuming a fixed response function. This phenomenon has been documented across multiple time scales: Hidalgo-Muñoz et al. (2018) found that heart rate decreased and HRV increased over the course of a single flight simulation due to habituation, effectively changing $P_t(y | x)$ within one session; Getzmann et al. (2021) reported that intra-individual test-retest correlations for EEG-based workload were low over 12–15 months, suggesting that $P_{t+\Delta}(y | x)$ can diverge substantially from $P_t(y | x)$ even when the same task is repeated; and Feltman (2016) demonstrated that circadian typology moderated subjective workload reports, indicating that $P_t(y | x)$ varies not only with task exposure but also with time of day. Standard supervised learning assumes a fixed mapping $P_t(y | x)$ that does not change over time; for MWL assessment, this assumption is violated by design, meaning that any model deployed without continuous recalibration will degrade in accuracy as the session progresses. Because this drift is directional within a session, a fixed model degrades monotonically after deployment rather than fluctuating randomly. Test–retest correlation analysis over intervals from days to months (Getzmann et al., 2021; Miyake

et al., 2009) is the standard method for assessing this reliability loss, and the magnitude of drift determines the required recalibration frequency.

(c) Emotional and Psychological States

Physiological measures of MWL are sensitive to emotional arousal and psychological states that may be unrelated to the cognitive task at hand. Hidalgo-Muñoz et al. (2018) manipulated emotion via a social stressor in a flight simulator and found that although emotional arousal enhanced motivation and performance, it was not reflected in cardiovascular activity, suggesting that physiological measures may fail to capture important psychological modulators. Emotional arousal can also contaminate other physiological signals that are interpreted as workload indices. Longo (2018) argued that transient effects of mood, motivation, or task outcomes on reported workload undermine the predictive validity of perceived workload relative to actual cognitive demand. Marchand et al. (2021) highlighted that physiological measures based on autonomic nervous system responses are fundamentally influenced by many factors beyond MWL, including physical activity and emotional state. The inability of physiological sensors to discriminate between cognitive load and emotional arousal represents a significant confound, particularly in applied settings where individuals may experience stress, anxiety, or frustration independently of task difficulty.

Causally, emotional arousal introduces a confounding path: emotional state E simultaneously influences both the observed physiology X (e.g., elevated heart rate due to anxiety) and the target construct Y (mental workload), creating the structure $X \leftarrow E \rightarrow Y$. Because E affects both variables, any observed correlation between X and Y cannot be uniquely attributed to the causal effect of workload on physiology; part of the observed association may instead reflect the confounding influence of emotional arousal. This structure violates the backdoor criterion for valid causal inference (Pearl, 2009), which requires that all backdoor paths from Y to X be blocked—either by conditioning on the confounder E or by design. In practical terms, unless the analysis explicitly controls for or conditions on emotional state, the estimated relationship between physiological signal and workload remains causally uninterpretable. The bias from this confound is directional under stress: anxious or frustrated individuals are flagged as high workload even when cognitive demand is moderate, producing false positives in exactly the situations where monitoring matters most. Concurrent measurement or manipulation of emotional state (Hidalgo-Muñoz et al., 2018) and disentangled representations (Section 4) are the candidate assessments.

(d) Expertise and Skill Level

The relationship between task demands and physiological response is mediated by expertise. Johnson et al. (2014) compared psychophysiological metrics between expert and novice performers in a deadly force judgment task and found that the same variables explained up to 72% of the variance in expert performance but only 37% in novice performance. Experts exhibited lower heart rate and different patterns of EEG alpha suppression than novices, indicating that physiological responses to workload are not equivalent across skill levels. Manning et al. (2006) reported that expert radiologists showed more efficient visual search strategies than novices when examining complex cases, suggesting that expertise alters both the subjective experience and the physiological cost of the same task. Waite et al. (2019) further demonstrated that expert radiologists fixate on abnormalities faster and with fewer total fixations than novices, reflecting different cognitive strategies that inevitably alter physiological signatures. These findings imply that a physiological model trained on novices may misestimate workload in experts, and vice versa, because the same objective task load elicits different internal states depending on the individual's skill level.

This violates the invariant prediction assumption underlying valid cross-population generalization: the conditional distribution $P(Y | X)$ —that is, the probability of workload state Y given a physiologically equivalent signal X under causal intervention—should remain stable across different subject populations (Peters et al., 2016). However, the physiological response function mapping signal to workload is expertise-dependent, denoted $f_{E(X)}$, where E represents expertise level. Because experts and novices recruit different cognitive

strategies and physiological resources to perform the same task (Johnson et al., 2014; Manning et al., 2006), the function $f_{E(X)}$ differs structurally between populations: the same physiological value X corresponds to different underlying workload states Y depending on whether the subject is an expert or a novice. Consequently, a model trained on one expertise population learns the wrong mapping when applied to another, as the causal structure relating physiology to workload changes across environments. The bias is directional and opposite across populations: a model trained on novices underestimates workload in experts, whereas one trained on experts overestimates it in novices. Expertise-stratified training and evaluation—or expertise entered as a covariate—provide both the assessment and the remedy (Section 4).

In summary, inter-individual variability, intra-individual temporal dynamics, emotional states, and expertise level all contribute noise and ambiguity to physiological MWL assessment. These factors operate independently of the technical quality of the measurement pipeline and cannot be fully controlled through improved sensors or algorithms. Matthews et al. (2015) emphasized that workload levels should not be determined from a single measure, especially in multitasking environments where different individuals may recruit different physiological resources. Future research must address these human-centered sources of variability through individualized calibration, multimodal fusion, and dynamic adaptation strategies that account for each individual's baseline state, circadian phase, and skill level.

4. Rethinking MWL Measurement

The analysis in Section 3 reveals that the problems of physiological MWL measurement are not merely technical but deeply structural. The current paradigm assumes that MWL is a stable, internally consistent state that can be inferred from physiological signals through a standardized pipeline. This assumption is contradicted by the evidence reviewed above: MWL is dynamic, multidimensional, and individually variable; physiological signals are multiplexed and context-sensitive; and the labels used for supervised learning are at best proxies and at worst misleading. If these structural problems are acknowledged, what directions might lead to more valid measurement? This section proposes a reconceptualization of the measurement problem and derives from it three interconnected directions for improvement: tightening the conditions under which current methods are applied, redesigning the measurement pipeline to preserve rather than discard uncertainty, and building models that adapt to human variability rather than ignore it.

4.1. When Can Current Methods Still Provide Useful Information?

The limitations identified in Section 3 do not render existing methods worthless. Rather, they delimit the conditions under which physiological signals can be interpreted as indicators of MWL with reasonable confidence. These conditions are not met in most published studies, but they are achievable in principle.

First, the measurement context must be controlled or at least monitored. Environmental factors such as ambient light, temperature, and humidity influence physiological signals independently of cognitive demands. Pupil diameter, for instance, is sensitive to illumination changes that have nothing to do with workload. If these factors cannot be held constant, they should be recorded and treated as covariates rather than ignored. A model that cannot determine whether a change in pupil size reflects increased cognitive load or a cloud passing overhead cannot produce valid inferences (Charles & Nixon, 2019; Kramer, 1990). Sciaraffa et al. (2022) illustrated the same concern: noting that skin conductance varies with environmental temperature and pupillometry with room luminance, they validated a light EEG-based measure for real-time stress monitoring under realistic driving conditions, showing that accounting for measurement context is not merely a theoretical recommendation but an empirically grounded practice.

Second, the task design must produce sufficiently distinct demand levels. If the difference between low- and high-workload conditions is subtle, physiological signals may not reliably distinguish them, and any observed classification accuracy may reflect noise or confounds rather than genuine workload sensitivity. The N-back task, for example, provides a relatively clean manipulation of working memory load, but even this can be confounded by individual differences in working memory capacity. Tasks with clearer demand gradients and stronger ecological validity are more likely to produce interpretable physiological signatures. Ayres et al. (2021) found that physiological measures were more sensitive to differences between memory and arithmetic tasks (90% detection rate) than to differences between simulations and object/shape manipulations (69% and 67%, respectively), demonstrating that task demand separation directly affects measurement sensitivity. Beh et al. (2021) likewise confirmed the value of the N-back paradigm as a benchmark for inducing graded MWL levels.

Third, within-subject designs should be preferred over between-subject designs when the goal is to track an individual's MWL over time. Individual differences in baseline physiology are substantial (see Section 3.1). A model trained on data pooled from multiple individuals assumes a common physiological signature of workload that the evidence does not support. Within-subject designs control for baseline differences by comparing each individual against their own reference states, though this comes at the cost of reduced generalizability. Tervonen et al. (2026) systematically evaluated personalization methods requiring only five minutes of baseline data across 170 participants, finding that continual normalization and baseline calibration with parameter prediction consistently outperform non-personalized models in multiclass cognitive state detection. Han et al. (2024) developed META-EEG, which combines gradient-based meta-learning with intermittent freezing strategies and outperformed baselines by 12% in zero-calibration motor imagery BCI systems. Ayres et al. (2021) found that within-subject designs produce more valid results because they control for prior knowledge and individual differences. Cai et al. (2026) demonstrated that few-shot cross-subject EEG cognitive load assessment can achieve competitive accuracy with only 3–5 labeled samples per class, suggesting that the generalizability cost of within-subject designs can be mitigated through meta-learned parameter initialization.

Fourth, multiple measures should be combined rather than relying on a single modality. Different physiological systems respond to different aspects of cognitive demand, and no single measure captures the full complexity of MWL. Cardiovascular measures reflect autonomic arousal, EEG measures reflect cortical activation, and eye-tracking measures reflect attention allocation. When these measures converge, confidence in the inference increases; when they diverge, the divergence itself is informative, signaling that the situation may be more complex than a single dimension can capture. Khan et al. (2024) proposed a convolutional neural network–long short-term memory (CNN-LSTM) architecture integrating fNIRS, eye-tracking, and vehicle dynamics data that achieved 99.82% accuracy in cognitive load prediction during driving simulations, demonstrating that multimodal fusion captures complementary spatial and temporal features. Planke et al. (2021) showed that online multimodal fusion of EEG, eye activity, and control inputs yields accurate and reliable inference of MWL in dynamic operational scenarios. Kosch et al. (2023), in a comprehensive survey of cognitive workload measurement in human–computer interaction (HCI), identified multimodal approaches combining EDA, EEG, and eye-tracking as the most promising direction for robust real-world assessment. Furthermore, Ding et al. (2025) proposed a multi-head cross-attention framework for multimodal physiological emotion recognition that achieves 94.88% valence and 95.26% arousal accuracy on DEAP, suggesting that attention-based fusion mechanisms can dynamically and effectively weight modality contributions.

Fifth, and most critically, the validity of the measurement must be evaluated against something other than task conditions. If the only criterion for validity is that the model correctly classifies low- versus high-difficulty trials, then the model has learned task discrimination, not workload assessment. Independent validation requires concurrent self-report ratings, performance indices, or, ideally, manipulations that affect experienced workload without changing task demands (or vice versa). Without such validation, high classification accuracy is scientifically hollow. Boring et al. (2020) demonstrated that continuous decoding of cognitive load from EEG

reveals both task-general and task-specific correlates, suggesting that validity assessment must distinguish between generic arousal patterns and demand-specific responses. Bhagat-Smith et al. (2024) proposed uncertainty-aware visual workload estimation, showing that models that provide confidence intervals rather than point estimates better identify when predictions are unreliable. Das Chakladar and Roy (2024), in their comprehensive review of cognitive workload estimation using physiological measures, concluded that concurrent validation against multiple independent criteria—self-report, performance, and physiological concordance—is essential for establishing construct validity.

These conditions are demanding, and many published studies fail to meet them. Yet they are not unattainable. The question is whether meeting them is sufficient, or whether more fundamental changes to the measurement paradigm are needed. Feltman (2016) found that circadian typology moderated subjective workload reports, suggesting that time-of-day effects must be accounted for even in well-controlled studies. Hidalgo-Muñoz et al. (2018) found that time-on-task effects changed cardiovascular signatures within a single session, demonstrating that baseline states are not stable reference points. Johnson et al. (2014) showed that the same physiological variables explained 72% of the variance in expert performance but only 37% in novice performance, underscoring that population characteristics fundamentally alter the signal-to-noise ratio.

4.2. Redesigning the Pipeline: From Deterministic Classification to Probabilistic Inference

Even when the conditions above are satisfied, the current measurement pipeline contains structural features that limit validity. The pipeline treats MWL as a discrete category to be classified, discards temporal dynamics through windowing and feature aggregation, and produces point estimates that conceal uncertainty. Each of these features can be redesigned (Prasetyo & Iridiastadi, 2025). Beyond these conceptual limitations, a lack of standardization across studies compounds the problem: variations in sensor placement, sampling rates, filter parameters, artifact rejection criteria, and feature computation methods make cross-study comparison difficult and may contribute to poor replicability (Pereira et al., 2025; Scrivener & Reader, 2022).

The problem of discrete labeling is perhaps the most fundamental. MWL is not a set of boxes labeled low, medium, and high; it is a continuous variable that fluctuates over time in response to task demands, environmental factors, and internal states. Cognitive resources are not allocated in discrete chunks. A discrete classification model imposes arbitrary boundaries on a continuous phenomenon, discarding information about the magnitude and trajectory of workload changes. Moreover, the relationship between physiological signals and MWL is not one-to-one but probabilistic: the same physiological pattern can arise from different psychological states, and the same MWL level can be realized through different physiological configurations. The more appropriate question is therefore not which category a person falls into, but what the probability distribution over possible workload levels is, given the observed data (Charles & Nixon, 2019).

Most current studies discretize MWL into two, three, or five classes (Grimaldi et al., 2024; Ji et al., 2023). Ji et al. (2023) built a continuous workload recognizer using dynamic residual networks with attention mechanisms, demonstrating that regression-based approaches preserve magnitude information and capture gradual workload transitions that classification methods miss. Grimaldi et al. (2024) proposed CNN-LSTM models for perceptual and cognitive load forecasting in piloting tasks, showing that LSTM-based approaches outperform discrete classifiers in short-term continuous workload prediction.

This reconceptualization has practical implications. Rather than outputting a single class label, a model could output a probability distribution over a continuous workload scale, where the mean represents the most likely workload level and the variance reflects the uncertainty surrounding that estimate. A wide, flat distribution indicates that the physiological data are consistent with multiple workload levels, signaling that the inference should be treated with caution, whereas a narrow, peaked distribution indicates clear signals and higher confidence. This approach makes uncertainty explicit rather than hiding it behind a deterministic output. Ozek

(2024) proposed neural network-based prediction intervals for pain estimation from EDA signals, demonstrating that quantifying both epistemic and aleatoric uncertainty yields more reliable inferences than point estimates. Ochella et al. (2024) applied Bayesian neural networks to remaining useful life prediction, showing that uncertainty-aware models provide confidence bounds that flag unreliable predictions. Tervonen et al. (2026) found that baseline calibration with parameter prediction—using a regressor to estimate individual normalization parameters from minimal baseline data—outperforms non-personalized models in multiclass cognitive state detection, suggesting that calibration efficiency can itself be learned and optimized. Fig. 2 illustrates this contrast: discrete classification imposes arbitrary cut-offs on overlapping condition distributions (Fig. 2a), whereas continuous probabilistic estimation outputs a distribution whose mean and variance make both the most likely workload level and the associated uncertainty explicit (Fig. 2b).

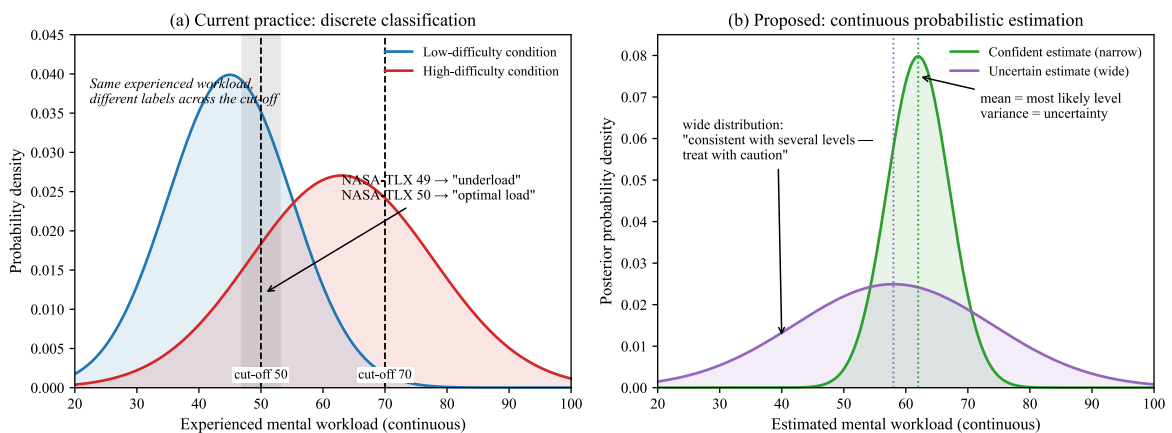


Fig. 2. From discrete classification to continuous probabilistic estimation. (a) Discrete cut-offs assign different labels to near-identical levels of experienced workload. (b) A continuous probabilistic estimate expresses both the most likely workload level (mean) and its uncertainty (variance). Distributions are illustrative.

The temporal dimension can be preserved rather than discarded. Current practice segments continuous signals into fixed windows and extracts summary statistics, thereby losing information about how workload evolves over time. Time-series models, recurrent architectures, or state-space approaches can instead model workload as a dynamic process governed by both task demands and internal state dynamics. This matters because workload is not purely stimulus-driven; it depends on the individual's prior state, expectations, and coping strategies, all of which evolve over the course of a session. Manawadu et al. (2018) applied LSTM-based recurrent neural networks to multiclass driver workload classification, demonstrating that temporal models capture sequential dependencies in physiological signals that static window-based approaches miss. Boring et al. (2020) showed that continuously decoding cognitive load from EEG reveals temporal dynamics invisible to trial-averaged classification, including anticipatory activation and recovery patterns. Khan et al. (2024) demonstrated that CNN-LSTM architectures integrating spatial and temporal feature extraction achieve 99.82% accuracy in driver workload prediction, suggesting that hybrid architectures are particularly well suited to capturing the spatiotemporal structure of workload-related physiological responses.

The feature extraction stage can be made less reductive. Hand-crafted features embody assumptions about which aspects of the physiological signal are relevant, and these assumptions may be wrong or incomplete. End-to-end learning can extract representations directly from raw or minimally processed signals, potentially capturing patterns that human-designed features miss. This does not eliminate the need for domain knowledge;

rather, it shifts the role of domain knowledge from dictating features to constraining the space of learnable representations (Roy et al., 2019). What can be retained from current approaches is the demonstrated ability of deep learning to capture non-linear relationships between physiological features and MWL-related variables. Zhang et al. (2024) showed that a convolutional neural network with an attention mechanism predicts pilot workload from fNIRS signals in flight simulations with 88.69% accuracy. This result demonstrates that the core capability—using machine learning to map physiological patterns to cognitive states—is sound; what needs to change is how this capability is deployed: from discrete deterministic classification to continuous probabilistic estimation, from static window-based features to dynamic temporal models, and from hand-crafted features to learned representations.

4.3. Addressing Human Variability: From Universal Models to Adaptive Systems

The technical improvements discussed above concern the pipeline, but they do not resolve the human-centric sources of variability identified in Section 3.7. Inter-individual differences, intra-individual temporal dynamics, emotional states, and expertise levels all modulate physiological responses in ways that standardized models cannot capture. Addressing these sources of variability requires a shift from universal models to adaptive systems.

For inter-individual variability, the solution is not to seek a single model that works for everyone, but to build models that can be quickly personalized. Meta-learning approaches learn a prior over possible response patterns from data across many individuals, enabling rapid adaptation to new individuals with minimal calibration data. Rather than assuming that all individuals share the same physiological signature, these approaches treat individual differences as a core design requirement. The amount of calibration data required becomes a measurable property of the model: a model that adapts to a new individual with only minutes of data is more practically useful than one that requires hours. Ng and Guan (2024) developed a few-shot subject-independent meta-learning framework that uses known subjects to prime the model for faster adaptation with only target samples during fine-tuning, achieving 88.70% accuracy in binary motor imagery classification without requiring extensive calibration. An et al. (2020) employed few-shot relation learning with attention for EEG-based motor imagery classification, demonstrating that relation networks can generalize to new subjects with minimal labeled data. In a comprehensive review of few-shot learning for EEG classification, Ahuja and Sethia (2024) confirmed that meta-learning approaches consistently outperform traditional transfer learning in cross-subject scenarios, with methods based on model-agnostic meta-learning (MAML) showing particular promise for rapid personalization. Collectively, these approaches reject the assumption of universal physiological signatures in favor of learned priors that can be quickly adapted to new individuals.

For intra-individual temporal dynamics, continuous adaptation mechanisms can track an individual's changing physiological baseline over the course of a session. Online test-time adaptation adjusts model parameters in real time based on incoming data, without requiring explicit recalibration or interrupting the primary task. This addresses time-on-task effects, habituation, and fatigue-related changes that would otherwise degrade model performance. The key challenge is to distinguish genuine baseline shifts from workload-related changes, a problem that may require additional information sources such as task context or behavioral markers. Wang et al. (2023) proposed optimization-free test-time adaptation (OFTTA) for cross-person activity recognition, eliminating the need for gradient computation during adaptation and making real-time personalization feasible on resource-constrained devices. Wang et al. (2021) proposed TENT (fully test-time adaptation by entropy minimization), which adapts pre-trained models to the test data distribution by minimizing the entropy of model predictions, achieving state-of-the-art performance without requiring labeled target data or access to source data. Tervonen et al. (2026) found that continual normalization—updating normalization parameters dynamically as new data arrives—outperforms static baseline calibration in free-

living conditions, suggesting that online adaptation is particularly well suited to applications where data accumulate gradually. These approaches address the core challenge of distinguishing baseline shifts from workload-related changes by treating adaptation as a continuous rather than discrete process.

For emotional and psychological confounds, disentangled representation learning offers a promising direction. The core idea is to learn separate latent representations for different psychological constructs, such that the workload representation is invariant to emotional arousal and vice versa. This requires either multi-task learning with multiple labeled constructs or unsupervised disentanglement techniques that exploit the statistical structure of the data. In practice, ground-truth labels for both workload and emotion are rarely available simultaneously, so semi-supervised or self-supervised approaches may be necessary. Zubair et al. (2024) proposed deep representation learning for multimodal emotion recognition using physiological signals, explicitly modeling the consistency and heterogeneity of EEG, ECG, and EDA through modality-invariant and modality-specific encoders. Yang et al. (2022) developed a disentangled representation learning framework for multimodal emotion recognition that separates shared emotional content from modality-specific noise, improving robustness to missing or corrupted modalities. Jia et al. (2024) proposed a Multi-level Disentangling Network (MDNet) for cross-subject emotion recognition based on multimodal physiological signals, separating modality-invariant from modality-specific features and subject-shared from subject-private features, and achieving state-of-the-art performance by modeling heterogeneity rather than treating it as noise. Applied to MWL measurement, these architectures suggest that workload-related variance can be separated from emotional arousal through hierarchical disentanglement, though obtaining the simultaneous labels for both constructs that this requires remains difficult in practice.

To account for expertise-related differences, models should either incorporate expertise level as a covariate or be trained and evaluated on expertise-stratified data. A model trained on novices may systematically underestimate workload in experts, who exhibit lower physiological arousal for the same task demands because of more efficient processing. Conversely, a model trained on experts may overestimate workload in novices. Explicitly modeling the expertise–workload interaction, or maintaining separate models for different expertise levels, would reduce such systematic errors. Hebbar et al. (2023) estimated pilots' cognitive load from ocular and EEG measures in a virtual reality (VR) flight simulator, showing that physiological workload metrics track perceived task difficulty in training settings. Bhumika et al. (2022) personalized stress estimation for taxi drivers and recommended trip decisions based on the driver's current physiological state, illustrating the value of adapting systems to individual differences. Sakib et al. (2022) predicted impending drone accidents from operators' physiological signals several seconds in advance, demonstrating that operator-state models can support real-time safety monitoring. For MWL assessment, these examples underscore the value of adapting models to the individual rather than assuming a homogeneous population.

These adaptive strategies are not mutually exclusive. A single framework could employ continuous probabilistic outputs, incorporate temporal dynamics, adapt online to baseline changes, disentangle emotion from workload, and condition on expertise level simultaneously. The technical feasibility of such frameworks is increasing, but their empirical validation remains limited. Contextual multi-task learning can jointly model workload alongside related constructs such as emotion, stress, and fatigue, exploiting their shared structure while allowing construct-specific variation. Rather than treating MWL as an isolated target, multi-task frameworks learn representations that disentangle overlapping psychological states, acknowledging that physiological signals are fundamentally multiplexed and that valid MWL measurement requires modeling the full psychological state space rather than a single dimension.

4.4. *What These Directions Do and Do Not Resolve*

The directions outlined above address many of the limitations identified in Section 3, but they do not resolve all of them. Continuous probabilistic outputs address the discrete labeling problem and make uncertainty explicit. Temporal modeling addresses the dynamics problem. Meta-learning and online adaptation address cross-individual and intra-individual variability. Disentangled representations address the emotion–workload confound. Expertise-aware modeling addresses population heterogeneity.

Momeni et al. (2019) implemented real-time cognitive workload monitoring using machine learning and physiological signals in rescue missions, finding that even multimodal systems require task-specific calibration to maintain validity across operational contexts. Dell'Agnola et al. (2021) developed MBioTracker, a multimodal self-aware bio-monitoring wearable system for online workload detection, demonstrating that integrated multi-sensor approaches improve robustness but still exhibit context-dependent variations in accuracy. These findings suggest that even combining all proposed methodological directions may not fully resolve the measurement problem without concurrent validation against ground-truth workload indicators.

However, the fundamental problem of construct validity remains. Any label—whether task difficulty, NASA-TLX score, or a learned latent variable—is only a proxy for experienced workload. A model trained to predict a proxy may predict that proxy accurately, but this does not guarantee that it measures the construct of interest. The gap between operational labels and theoretical constructs is as much a philosophical problem as a technical one, and no amount of methodological sophistication can eliminate it entirely. Hidalgo-Muñoz et al. (2018) found that emotional arousal can enhance performance without altering cardiovascular activity, which means that context recording may not always disambiguate cognitive load from emotional state. This illustrates the deeper problem: even when models are technically sophisticated, the labels they predict may not correspond to the psychological construct of interest. In a comprehensive survey of cognitive workload measurement in HCI, Kosch et al. (2023) identified the fundamental tension between operational definitions (task performance) and experiential definitions (subjective reports) as a persistent barrier to construct validity that methodological improvements alone cannot resolve.

Moreover, the adaptive strategies described above assume that the relevant individual characteristics can be identified and measured. Yet some sources of variability may be unobservable or only partially observable. A person's motivational state, current concerns, expectations about the task, and momentary mood all influence physiological responses, yet none of these are easily measured in real time. Even the best model cannot adjust for variables it does not know about. Yet even the minimal-data personalization approaches evaluated by Tervonen et al. (2026) cannot capture momentary motivational states, task expectations, or transient mood fluctuations, which remain difficult to measure in real time and represent a fundamental limit to adaptive modeling.

These limitations suggest that the goal should not be to eliminate all sources of variability and produce a perfect measurement instrument. Such an instrument is probably impossible in principle, given the complexity of human cognition and physiology. A more realistic goal is to build systems that are transparent about their limitations, adaptive to known sources of variability, and validated against multiple independent criteria. The value of physiological MWL measurement lies not in achieving perfect accuracy but in providing continuous, objective information that complements rather than replaces subjective and performance-based measures.

4.5. *Summary*

This section has proposed a reconceptualization of physiological MWL measurement grounded in the critical analysis in Section 3. Rather than treating MWL as a discrete category to be classified from standardized signals, the proposed framework treats it as a continuous, dynamic, probabilistic variable inferred from multiple

measures under controlled conditions. The framework recognizes that human variability is not noise to be eliminated but information to be incorporated. It demands that validity be established against independent criteria rather than inferred from classification accuracy alone. It also acknowledges that even the most sophisticated model cannot fully bridge the gap between physiological signals and subjective experience.

These principles translate into a concrete measurement protocol that addresses the limitations identified in this paper. First, the measurement session should begin with a standardized baseline recording under controlled environmental conditions, with ambient light, temperature, and noise held constant or recorded as covariates. This baseline serves as the individual reference against which subsequent deviations are evaluated, not as a universal norm. Second, the primary task should be designed with sufficiently distinct demand levels, verified through pilot testing to ensure that physiological signals can reliably distinguish them. Tasks with subtle demand gradients should be avoided, as they produce ambiguous physiological signatures that confound measurement validity. Third, multiple physiological modalities should be recorded simultaneously, with particular attention to their complementary contributions: EEG for cortical activation, EDA for sympathetic arousal, and eye-tracking for attention allocation. The convergence and divergence across these measures should be analyzed jointly rather than independently.

Fourth, the analytical model should treat MWL as a continuous variable and output probability distributions rather than discrete labels, making uncertainty explicit and flagging ambiguous inferences for human review. Temporal dynamics should be preserved through recurrent or state-space architectures rather than discarded through fixed-window aggregation. Fifth, the model should be personalized to each individual through meta-learning or online adaptation, using minimal calibration data collected at session onset. The efficiency of this personalization—the amount of calibration data required for acceptable accuracy—should be reported as a key metric of model quality. Sixth, expertise level and emotional state should be assessed concurrently and incorporated as covariates or stratification variables, not treated as nuisance factors to be ignored. Finally, and most critically, the validity of MWL estimates should be evaluated against concurrent self-report ratings and performance indices, with mismatches between these criteria treated as diagnostic information rather than errors to be minimized.

This protocol does not guarantee perfect measurement. Construct validity remains fundamentally limited by the gap between operational labels and subjective experience, and some sources of individual variability are inherently unobservable. However, by systematically addressing the conditions of measurement, the structure of the inference pipeline, and the adaptive modeling of human variability, this protocol defines the conditions under which physiological MWL measurement can achieve acceptable validity for research and applied purposes. The goal is not to eliminate uncertainty but to make it transparent, manageable, and informative.

5. Conclusion

This paper has critically examined current physiological and machine learning approaches to mental workload assessment. Although many studies report high classification accuracy, several methodological limitations persist across the entire pipeline, from signal acquisition and preprocessing to feature extraction, labeling, and model evaluation. Physiological responses are influenced by multiple factors beyond MWL, and current supervised learning approaches often rely on imperfect proxy labels. Moreover, treating MWL as discrete categories oversimplifies a fundamentally continuous and dynamic cognitive process. These concerns are not merely theoretical; they have direct implications for the validity and generalizability of MWL assessment in safety-critical applications.

Rather than rejecting existing approaches, this paper has advocated a more nuanced application of current methods under specified conditions, together with a conceptual shift toward continuous and probabilistic MWL estimation. Future research should prioritize construct validity over classification accuracy, develop

standardized protocols for signal acquisition and processing, and pursue methodological innovations that address individual variability, temporal dynamics, and label validity. The integration of multimodal physiological signals offers a foundation for more robust assessment, but only when coupled with methodological rigor and theoretical clarity. Ultimately, the goal is not perfect prediction, but to ensure that model outputs faithfully reflect human cognitive states while maintaining appropriate humility about the limitations of physiological measurement.

Acknowledgements

References

- Adams, C. E., & Leverland, M. B. (1985). Environmental and Behavioral Factors That Can Affect Blood Pressure. *The Nurse Practitioner*, 10(11), 39. <https://doi.org/10.1097/00006205-198511000-00005>
- Aghajani, H., Garbey, M., & Omurtag, A. (2017). Measuring Mental Workload with EEG+fNIRS. *Frontiers in Human Neuroscience*, 11, 359. <https://doi.org/10.3389/fnhum.2017.00359>
- Ahuja, C., & Sethia, D. (2024). Harnessing Few-Shot Learning for EEG signal classification: A survey of state-of-the-art techniques and future directions. *Frontiers in Human Neuroscience*, 18, 1421922. <https://doi.org/10.3389/fnhum.2024.1421922>
- Aksu, Ş. H., Çakıt, E., & Dağdeviren, M. (2024). Mental Workload Assessment Using Machine Learning Techniques Based on EEG and Eye Tracking Data. *Applied Sciences*, 14(6), 2282. <https://doi.org/10.3390/app14062282>
- An, S., Kim, S., Chikontwe, P., & Park, S. H. (2020). Few-Shot Relation Learning with Attention for EEG-based Motor Imagery Classification. 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 10933–10938. <https://doi.org/10.1109/IROS45743.2020.9340933>
- Ayres, P., Lee, J. Y., Paas, F., & Van Merriënboer, J. J. G. (2021). The Validity of Physiological Measures to Identify Differences in Intrinsic Cognitive Load. *Frontiers in Psychology*, 12, 702538. <https://doi.org/10.3389/fpsyg.2021.702538>
- Backs, R. W. (1995). Going Beyond Heart Rate: Autonomic Space and Cardiovascular Assessment of Mental Workload. *The International Journal of Aviation Psychology*, 5(1), 25–48. https://doi.org/10.1207/s15327108ijap0501_3
- Beh, W.-K., Wu, Y.-H., An-Yeu, & Wu. (2021). MAUS: A Dataset for Mental Workload Assessment on N-back Task Using Wearable Sensor (arXiv:2111.02561). arXiv. <https://doi.org/10.48550/arXiv.2111.02561>
- Ben-David, S., Blitzer, J., Crammer, K., Kulesza, A., Pereira, F., & Vaughan, J. W. (2010). A theory of learning from different domains. *Machine Learning*, 79(1–2), 151–175. <https://doi.org/10.1007/s10994-009-5152-4>
- Bernardi, L., Wdowczyk-Szulc, J., Valenti, C., Castoldi, S., Passino, C., Spadacini, G., & Sleight, P. (2000). Effects of controlled breathing, mental activity and mental stress with or without verbalization on heart rate variability. *Journal of the American College of Cardiology*, 35(6), 1462–1469. [https://doi.org/10.1016/S0735-1097\(00\)00595-7](https://doi.org/10.1016/S0735-1097(00)00595-7)
- Bhagat Smith, J., Angelo Toribio, S., & Adams, J. (2024). Uncertainty-Aware Visual Workload Estimation for Human-Robot Teams. 126–111. <https://doi.org/10.29007/6w5h>
- Bhumika, Vyas, J., & Das, D. (2022). Stress-Aware Recommendation for Safe Driving using MTL-ConvLSTM. 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC), 3490–3495. <https://doi.org/10.1109/ITSC55140.2022.9922291>
- Boring, M. J., Ridgeway, K., Shvartsman, M., & Jonker, T. R. (2020). Continuous decoding of cognitive load from electroencephalography reveals task-general and task-specific correlates. *Journal of Neural Engineering*, 17(5), 056016. <https://doi.org/10.1088/1741-2552/abb9bc>
- Borsboom, D., Mellenbergh, G. J., & Van Heerden, J. (2004). The Concept of Validity. *Psychological Review*, 111(4), 1061–1071. <https://doi.org/10.1037/0033-295X.111.4.1061>
- Brookings, J. B., Wilson, G. F., & Swain, C. R. (1996). Psychophysiological responses to changes in workload during simulated air traffic control. *Biological Psychology*, 42(3), 361–377. [https://doi.org/10.1016/0301-0511\(95\)05167-8](https://doi.org/10.1016/0301-0511(95)05167-8)
- Cai, R., Dai, S., Wu, X., Song, X., Li, M., & Hu, D. (2026). Few-Shot Cross-Subject EEG Cognitive Load Assessment Based on Global Cross-Attention Domain Adaptation. *IEEE Sensors Journal*, 26(10), 14813–14830. <https://doi.org/10.1109/JSEN.2025.3617004>
- Cárdenas-Vélez, D., Perales, J. C., Chiroso, L. J., Conde-González, J., Aguilar-Martínez, D., & Araya, S. (2013). The effect of mental workload on the intensity and emotional dynamics of perceived exertion. *Anales de Psicología*, 29(3), 662–673. <https://doi.org/10.6018/analesps.29.3.175801>
- Causse, M., Chua, Z., Peysakhovich, V., Del Campo, N., & Matton, N. (2017). Mental workload and neural efficiency quantified in the prefrontal cortex using fNIRS. *Scientific Reports*, 7(1), 5222. <https://doi.org/10.1038/s41598-017-05378-x>

- Causse, M., Sénard, J.-M., Démonet, J. F., & Pastor, J. (2010). Monitoring Cognitive and Emotional Processes Through Pupil and Cardiac Response During Dynamic Versus Logical Task. *Applied Psychophysiology and Biofeedback*, 35(2), 115–123. <https://doi.org/10.1007/s10484-009-9115-0>
- Charles, R. L., & Nixon, J. (2019). Measuring mental workload using physiological measures: A systematic review. *Applied Ergonomics*, 74, 221–232. <https://doi.org/10.1016/j.apergo.2018.08.028>
- Cheema, B. S., Samima, S., Sarma, M., & Samanta, D. (2018). Mental Workload Estimation from EEG Signals Using Machine Learning Algorithms. In D. Harris (Ed.), *Engineering Psychology and Cognitive Ergonomics* (pp. 265–284). Springer International Publishing. https://doi.org/10.1007/978-3-319-91122-9_23
- Collet, C., Salvia, E., & Petit-Boulanger, C. (2014). Measuring workload with electrodermal activity during common braking actions. *Ergonomics*, 57(6), 886–896. <https://doi.org/10.1080/00140139.2014.899627>
- Cover, T. M., & Thomas, J. A. (2006). *Elements of information theory second edition solutions to problems*. Wiley-Interscience.
- Das Chakladar, D., & Roy, P. P. (2024). Cognitive workload estimation using physiological measures: A review. *Cognitive Neurodynamics*, 18(4), 1445–1465. <https://doi.org/10.1007/s11571-023-10051-3>
- De Rivecourt, M., Kuperus, M. N., Post, W. J., & Mulder, L. J. M. (2008). Cardiovascular and eye activity measures as indices for momentary changes in mental effort during simulated flight. *Ergonomics*, 51(9), 1295–1319. <https://doi.org/10.1080/00140130802120267>
- Debie, E., Fernandez Rojas, R., Fidock, J., Barlow, M., Kasmarik, K., Anavatti, S., Garratt, M., & Abbass, H. A. (2021). Multimodal Fusion for Objective Assessment of Cognitive Workload: A Review. *IEEE Transactions on Cybernetics*, 51(3), 1542–1555. <https://doi.org/10.1109/TCYB.2019.2939399>
- Dell'Agnola, F., Pale, U., Marino, R., Arza, A., & Atienza, D. (2021). MBioTracker: Multimodal Self-Aware Bio-Monitoring Wearable System for Online Workload Detection. *IEEE Transactions on Biomedical Circuits and Systems*, 15(5), 994–1007. <https://doi.org/10.1109/TBCAS.2021.3110317>
- Di Flumeri, G., Borghini, G., Aricò, P., Sciaraffa, N., Lanzi, P., Pozzi, S., Vignali, V., Lantieri, C., Bichicchi, A., Simone, A., & Babiloni, F. (2018). EEG-Based Mental Workload Neurometric to Evaluate the Impact of Different Traffic and Road Conditions in Real Driving Settings. *Frontiers in Human Neuroscience*, 12, 509. <https://doi.org/10.3389/fnhum.2018.00509>
- Diarra, M., Theurel, J., & Paty, B. (2025). Systematic review of neurophysiological assessment techniques and metrics for mental workload representation in real-world settings. *Frontiers in Neuroergonomics*, 6, 1584736. <https://doi.org/10.3389/fnrgo.2025.1584736>
- Diaz-Piedra, C., Sebastián, M. V., & Di Stasi, L. L. (2020). EEG Theta Power Activity Reflects Workload among Army Combat Drivers: An Experimental Study. *Brain Sciences*, 10(4), 199. <https://doi.org/10.3390/brainsci10040199>
- Ding, S., Ma, L., & Li, H. (2025). Multimodal physiological signal emotion recognition based on multi-head cross attention with representation learning. *Frontiers in Psychiatry*, 16, 1713559. <https://doi.org/10.3389/fpsy.2025.1713559>
- Ding, Y., Cao, Y., Duffy, V. G., Wang, Y., & Zhang, X. (2020). Measurement and identification of mental workload during simulated computer tasks with multimodal methods and machine learning. *Ergonomics*, 63(7), 896–908. <https://doi.org/10.1080/00140139.2020.1759699>
- Fairclough, S. H., & Venables, L. (2006). Prediction of subjective states from psychophysiology: A multivariate approach. *Biological Psychology*, 71(1), 100–110. <https://doi.org/10.1016/j.biopsycho.2005.03.007>
- Fairclough, S. H., Venables, L., & Tattersall, A. (2005). The influence of task demand and learning on the psychophysiological response. *International Journal of Psychophysiology*, 56(2), 171–184. <https://doi.org/10.1016/j.ijpsycho.2004.11.003>
- Fallahi, M., Motamedzade, M., Heidarimoghadam, R., Soltanian, A. R., & Miyake, S. (2016). Effects of mental workload on physiological and subjective responses during traffic density monitoring: A field study. *Applied Ergonomics*, 52, 95–103. <https://doi.org/10.1016/j.apergo.2015.07.009>
- Fawwaz, A. A., Rahma, O. N., Ain, K., Ittaqillah, S. I., & Chai, R. (2024). Measurement of Mental Workload Using Heart Rate Variability and Electrodermal Activity. *IEEE Access*, 12, 197589–197601. <https://doi.org/10.1109/ACCESS.2024.3521649>
- Feltman, K. (2016). The effects of time of day and circadian rhythm on performance during variable levels of cognitive workload. Theses and Dissertations. <https://commons.und.edu/theses/348>
- Fournier, L. R., Wilson, G. F., & Swain, C. R. (1999). Electrophysiological, behavioral, and subjective indexes of workload when performing multiple tasks: Manipulations of task difficulty and training. *International Journal of Psychophysiology*, 31(2), 129–145. [https://doi.org/10.1016/S0167-8760\(98\)00049-X](https://doi.org/10.1016/S0167-8760(98)00049-X)
- Frey, J., Mühl, C., Lotte, F., & Hachet, M. (2014). Review of the Use of Electroencephalography as an Evaluation Method for Human-Computer Interaction: Proceedings of the International Conference on Physiological Computing Systems, 214–223. <https://doi.org/10.5220/0004708102140223>
- Gao, Q., Wang, Y., Song, F., Li, Z., & Dong, X. (2013). Mental workload measurement for emergency operating procedures in digital nuclear power plants. *Ergonomics*, 56(7), 1070–1085. <https://doi.org/10.1080/00140139.2013.790483>
- Getzmann, S., Reiser, J. E., Karthaus, M., Rudinger, G., & Wascher, E. (2021). Measuring Correlates of Mental Workload During Simulated Driving Using cEEGrid Electrodes: A Test–Retest Reliability Analysis. *Frontiers in Neuroergonomics*, 2, 729197. <https://doi.org/10.3389/fnrgo.2021.729197>
- Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B., & Smola, A. (2012). A Kernel Two-Sample Test. 13(25), 723–773. <https://doi.org/http://jmlr.org/papers/v13/gretton12a.html>
- Grimaldi, N., Liu, Y., McKendrick, R., Ruiz, J., & Kaber, D. (2024). Deep Learning Forecast of Cognitive Workload Using fNIRS Data. 2024 IEEE 4th International Conference on Human-Machine Systems (ICHMS), 1–6. <https://doi.org/10.1109/ICHMS59971.2024.10555701>

- Grimes, D., Tan, D. S., Hudson, S. E., Shenoy, P., & Rao, R. P. N. (2008). Feasibility and pragmatics of classifying working memory load with an electroencephalograph. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 835–844. <https://doi.org/10.1145/1357054.1357187>
- Guan, K., Zhang, Z., Liu, T., & Niu, H. (2023). Cross-Task Mental Workload Recognition Based on EEG Tensor Representation and Transfer Learning. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 31, 2632–2639. <https://doi.org/10.1109/TNSRE.2023.3277867>
- Hamann, A., & Carstengerdes, N. (2022). Investigating mental workload-induced changes in cortical oxygenation and frontal theta activity during simulated flights. *Scientific Reports*, 12(1), 6449. <https://doi.org/10.1038/s41598-022-10044-y>
- Han, J.-W., Bak, S., Kim, J.-M., Choi, W., Shin, D.-H., Son, Y.-H., & Kam, T.-E. (2024). META-EEG: Meta-learning-based class-relevant EEG representation learning for zero-calibration brain–computer interfaces. *Expert Systems with Applications*, 238, 121986. <https://doi.org/10.1016/j.eswa.2023.121986>
- Hebbar, A., Vinod, S., A.K., S., Pashilkar, A., & Biswas, P. (2023). Cognitive load estimation in VR flight simulator. *Journal of Eye Movement Research*, 15(3). <https://doi.org/10.16910/jemr.15.3.11>
- Hidalgo-Muñoz, A. R., Mouratille, D., Matton, N., Causse, M., Rouillard, Y., & El-Yagoubi, R. (2018). Cardiovascular correlates of emotional state, cognitive workload and time-on-task effect during a realistic flight simulation. *International Journal of Psychophysiology*, 128, 62–69. <https://doi.org/10.1016/j.ijpsycho.2018.04.002>
- Hogervorst, M. A., Brouwer, A.-M., & Van Erp, J. B. F. (2014). Combining and comparing EEG, peripheral physiology and eye-related measures for the assessment of mental workload. *Frontiers in Neuroscience*, 8. <https://doi.org/10.3389/fnins.2014.00322>
- Hwang, S.-L., Yau, Y.-J., Lin, Y.-T., Chen, J.-H., Huang, T.-H., Yenn, T.-C., & Hsu, C.-C. (2008). Predicting work performance in nuclear power plants. *Safety Science*, 46(7), 1115–1124. <https://doi.org/10.1016/j.ssci.2007.06.005>
- Iqbal, M. U., Srinivasan, B., & Srinivasan, R. (2024). Multi-class classification of control room operators' cognitive workload using the fusion of eye-tracking and electroencephalography. *Computers & Chemical Engineering*, 181, 108526. <https://doi.org/10.1016/j.compchemeng.2023.108526>
- Itoh, M. (2009). Individual differences in effects of secondary cognitive activity during driving on temperature at the nose tip. 2009 International Conference on Mechatronics and Automation, 7–11. <https://doi.org/10.1109/ICMA.2009.5246188>
- Ji, Z., Tang, J., Wang, Q., Xie, X., Liu, J., & Yin, Z. (2023). Cross-task cognitive workload recognition using a dynamic residual network with attention mechanism based on neurophysiological signals. *Computer Methods and Programs in Biomedicine*, 230, 107352. <https://doi.org/10.1016/j.cmpb.2023.107352>
- Jia, Z., Zhao, F., Guo, Y., Chen, H., & Jiang, T. (2024). Multi-level Disentangling Network for Cross-Subject Emotion Recognition Based on Multimodal Physiological Signals. *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*, 3069–3077. <https://doi.org/10.24963/ijcai.2024/340>
- Johnson, R. R., Stone, B. T., Miranda, C. M., Vila, B., James, L., James, S. M., Rubio, R. F., & Berka, C. (2014). Identifying psychophysiological indices of expert vs. Novice performance in deadly force judgment and decision making. *Frontiers in Human Neuroscience*, 8. <https://doi.org/10.3389/fnhum.2014.00512>
- Kahneman, D. (1973). *Attention and Effort*. Prentice-Hall.
- Kajiwara, S. (2014). Evaluation of driver's mental workload by facial temperature and electrodermal activity under simulated driving conditions. *International Journal of Automotive Technology*, 15(1), 65–70. <https://doi.org/10.1007/s12239-014-0007-9>
- Khan, M. A., Asadi, H., Qazani, M. R. C., Lim, C. P., & Nahavandi, S. (2024). Functional near-infrared spectroscopy (fNIRS) and Eye tracking for Cognitive Load classification in a Driving Simulator Using Deep Learning (Version 3). *arXiv*. <https://doi.org/10.48550/ARXIV.2408.06349>
- Kingphai, K., & Moshfeghi, Y. (2025). Mental Workload Assessment Using Deep Learning Models From EEG Signals: A Systematic Review. *IEEE Transactions on Cognitive and Developmental Systems*, 17(1), 40–60. <https://doi.org/10.1109/TCDS.2024.3460750>
- Kosch, T., Karolus, J., Zagermann, J., Reiterer, H., Schmidt, A., & Woźniak, P. W. (2023). A Survey on Measuring Cognitive Workload in Human-Computer Interaction. *ACM Computing Surveys*, 55(13s), 1–39. <https://doi.org/10.1145/3582272>
- Kramer, A. F. (1990). Physiological metrics of mental workload: A review of recent progress. In D. L. Damos (Ed.), *Multiple-task performance* (1st ed., pp. 279–328). CRC Press. <https://doi.org/10.1201/9781003069447-14>
- Kumar Das, U., Chowdhury, M., Liu, Y., & Kaber, D. (2025). A Systematic Review of Ground Truth Labeling and Prediction for Cognitive Workload Adaptive Systems. *AHFE HAWAII International Conference (AHFE HAWAII 2025)*. <https://doi.org/10.54941/ahfe1006883>
- Lean, Y., & Shan, F. (2012). Brief review on physiological and biochemical evaluations of human mental workload. *Human Factors and Ergonomics in Manufacturing & Service Industries*, 22(3), 177–187. <https://doi.org/10.1002/hfm.20269>
- Lee, C., Shin, M., Eniyandunmo, D., Anwar, A., Kim, E., Kim, K., Yoo, J. K., & Lee, C. (2024). Predicting Driver's mental workload using physiological signals: A functional data analysis approach. *Applied Ergonomics*, 118, 104274. <https://doi.org/10.1016/j.apergo.2024.104274>
- Lin, C. J., & Lukodono, R. P. (2022). Classification of mental workload in Human-robot collaboration using machine learning based on physiological feedback. *Journal of Manufacturing Systems*, 65, 673–685. <https://doi.org/10.1016/j.jmsy.2022.10.017>
- Longo, L. (2018). Experienced mental workload, perception of usability, their interaction and impact on task performance. *PLOS ONE*, 13(8), e0199661. <https://doi.org/10.1371/journal.pone.0199661>
- Longo, L., Wickens, C. D., Hancock, G., & Hancock, P. A. (2022). Human Mental Workload: A Survey and a Novel Inclusive Definition. *Frontiers in Psychology*, 13, 883321. <https://doi.org/10.3389/fpsyg.2022.883321>

- Manawadu, U. E., Kawano, T., Murata, S., Kamezaki, M., Muramatsu, J., & Sugano, S. (2018). Multiclass Classification of Driver Perceived Workload Using Long Short-Term Memory based Recurrent Neural Network. 2018 IEEE Intelligent Vehicles Symposium (IV), 1–6. <https://doi.org/10.1109/IVS.2018.8500410>
- Manning, D., Ethell, S., Donovan, T., & Crawford, T. (2006). How do radiologists do it? The influence of experience and training on searching for chest nodules. *Radiography*, 12(2), 134–142. <https://doi.org/10.1016/j.radi.2005.02.003>
- Marchand, C., De Graaf, J. B., & Jarrassé, N. (2021). Measuring mental workload in assistive wearable devices: A review. *Journal of NeuroEngineering and Rehabilitation*, 18(1), 160. <https://doi.org/10.1186/s12984-021-00953-w>
- Marchitto, M., Benedetto, S., Baccino, T., & Cañas, J. J. (2016). Air traffic control: Ocular metrics reflect cognitive complexity. *International Journal of Industrial Ergonomics*, 54, 120–130. <https://doi.org/10.1016/j.ergon.2016.05.010>
- Matthews, G., Reinerman-Jones, L. E., Barber, D. J., & Abich, J. (2015). The Psychometrics of Mental Workload: Multiple Measures Are Sensitive but Divergent. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 57(1), 125–143. <https://doi.org/10.1177/0018720814539505>
- Mehler, B., Reimer, B., Coughlin, J. F., & Dusek, J. A. (2009). Impact of Incremental Increases in Cognitive Workload on Physiological Arousal and Performance in Young Adult Drivers. *Transportation Research Record: Journal of the Transportation Research Board*, 2138(1), 6–12. <https://doi.org/10.3141/2138-02>
- Melnik, A., Legkov, P., Izdebski, K., Kärcher, S. M., Hairston, W. D., Ferris, D. P., & König, P. (2017). Systems, Subjects, Sessions: To What Extent Do These Factors Influence EEG Data? *Frontiers in Human Neuroscience*, 11. <https://doi.org/10.3389/fnhum.2017.00150>
- Miyake, S., Yamada, S., Shoji, T., Takae, Y., Kuge, N., & Yamamura, T. (2009). Physiological responses to workload change. A test/retest examination. *Applied Ergonomics*, 40(6), 987–996. <https://doi.org/10.1016/j.apergo.2009.02.005>
- Mohanavelu, K., Poonguzhali, S., Janani, A., & Vinutha, S. (2022). Machine learning-based approach for identifying mental workload of pilots. *Biomedical Signal Processing and Control*, 75, 103623. <https://doi.org/10.1016/j.bspc.2022.103623>
- Momeni, N., Dell’Agnola, F., Arza, A., & Atienza, D. (2019). Real-Time Cognitive Workload Monitoring Based on Machine Learning Using Physiological Signals in Rescue Missions. 2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 3779–3785. <https://doi.org/10.1109/EMBC.2019.8857501>
- Moustafa, K., Luz, S., & Longo, L. (2017). Assessment of Mental Workload: A Comparison of Machine Learning Methods and Subjective Assessment Techniques. In L. Longo & M. C. Leva (Eds.), *Human Mental Workload: Models and Applications* (Vol. 726, pp. 30–50). Springer International Publishing. https://doi.org/10.1007/978-3-319-61061-0_3
- Murai, K., & Hayashi, Y. (2008). An evaluation of mental workload for effective navigation. *Interactive Technology and Smart Education*, 5(1), 29–38. <https://doi.org/10.1108/17415650810871565>
- Ng, H. W., & Guan, C. (2024). Subject-independent meta-learning framework towards optimal training of EEG-based classifiers. *Neural Networks*, 172, 106108. <https://doi.org/10.1016/j.neunet.2024.106108>
- Nickel, P., & Nachreiner, F. (2003). Sensitivity and Diagnosticity of the 0.1-Hz Component of Heart Rate Variability as an Indicator of Mental Workload. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 45(4), 575–590. <https://doi.org/10.1518/hfes.45.4.575.27094>
- Ochella, S., Dinmohammadi, F., & Shafiee, M. (2024). Bayesian neural networks for uncertainty quantification in remaining useful life prediction of systems with sensor monitoring. *Advances in Mechanical Engineering*, 16(7), 16878132241239802. <https://doi.org/10.1177/16878132241239802>
- Ozek, B. (2024). Uncertainty quantification in pain assessment through machine learning [Northeastern University]. <https://doi.org/10.17760/D20659778>
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.
- Peng, H., Long, F., & Ding, C. (2005). Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 27(8), 1226–1238. <https://doi.org/10.1109/TPAMI.2005.159>
- Pereira, E., Sigcha, L., Silva, E., Sampaio, A., Costa, N., & Costa, N. (2025). Capturing Mental Workload Through Physiological Sensors in Human–Robot Collaboration: A Systematic Literature Review. *Applied Sciences*, 15(6), 3317. <https://doi.org/10.3390/app15063317>
- Peters, J., Bühlmann, P., & Meinshausen, N. (2016). Causal Inference by using Invariant Prediction: Identification and Confidence Intervals. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 78(5), 947–1012. <https://doi.org/10.1111/rssb.12167>
- Planke, L. J., Gardi, A., Sabatini, R., Kistan, T., & Ezer, N. (2021). Online Multimodal Inference of Mental Workload for Cognitive Human Machine Systems. *Computers*, 10(6), 81. <https://doi.org/10.3390/computers10060081>
- Prasetyo, R. A. B., & Iridiastadi, H. (2025). Mental Workload: Definition and Measurement Review. *International Journal of Technology*, 16(5), 1854–1876. <https://doi.org/10.14716/ijtech.v16i5.7072>
- Roy, Y., Banville, H., Albuquerque, I., Gramfort, A., Falk, T. H., & Faubert, J. (2019). Deep learning-based electroencephalography analysis: A systematic review. *Journal of Neural Engineering*, 16(5), 051001. <https://doi.org/10.1088/1741-2552/ab260c>
- Sakib, M. N., Chaspari, T., & Behzadan, A. H. (2022). A feedforward neural network for drone accident prediction from physiological signals. *Smart and Sustainable Built Environment*, 11(4), 1017–1041. <https://doi.org/10.1108/SASBE-12-2020-0181>
- Sciaraffa, N., Di Flumeri, G., Germano, D., Giorgi, A., Di Florio, A., Borghini, G., Vozzi, A., Ronca, V., Varga, R., Van Gasteren, M., Babiloni, F., & Arico, P. (2022). Validation of a Light EEG-Based Measure for Real-Time Stress Monitoring during Realistic Driving. *Brain Sciences*, 12(3), 304. <https://doi.org/10.3390/brainsci12030304>

- Sciaraffa, N., Germano, D., Giorgi, A., Ronca, V., Vozzi, A., Borghini, G., Di Flumeri, G., Babiloni, F., & Arico, P. (2021). Mental Effort Estimation by Passive BCI: A Cross-Subject Analysis. 2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), 906–909. <https://doi.org/10.1109/EMBC46164.2021.9630613>
- Scrivener, C. L., & Reader, A. T. (2022). Variability of EEG electrode positions and their underlying brain regions: Visualizing gel artifacts from a simultaneous EEG-fMRI dataset. *Brain and Behavior*, 12(2), e2476. <https://doi.org/10.1002/brb3.2476>
- Shannon, C. E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*, 27(3), 379–423. <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>
- Singh, G., Chandel, C. P. C., & Roy, R. N. (2021). Mental Workload Estimation Based on Physiological Features for Pilot-UAV Teaming Applications. *Frontiers in Human Neuroscience*, 15, 692878. <https://doi.org/10.3389/fnhum.2021.692878>
- Solis-Marcos, I., & Kircher, K. (2019). Event-related potentials as indices of mental workload while using an in-vehicle information system. *Cognition, Technology & Work*, 21(1), 55–67. <https://doi.org/10.1007/s10111-018-0485-z>
- Stapel, J., Mullakkal-Babu, F. A., & Happee, R. (2019). Automated driving reduces perceived workload, but monitoring causes higher cognitive load than manual driving. *Transportation Research Part F: Traffic Psychology and Behaviour*, 60, 590–605. <https://doi.org/10.1016/j.trf.2018.11.006>
- Subasi, A. (2005). Automatic recognition of alertness level from EEG by using neural network and wavelet coefficients. *Expert Systems with Applications*, 28(4), 701–711. <https://doi.org/10.1016/j.eswa.2004.12.027>
- Tao, D., Tan, H., Wang, H., Zhang, X., Qu, X., & Zhang, T. (2019). A Systematic Review of Physiological Measures of Mental Workload. *International Journal of Environmental Research and Public Health*, 16(15), 2716. <https://doi.org/10.3390/ijerph16152716>
- Tervonen, J., Amparore, E. G., Botta, M., Närviäinen, J., Pettersson, K., & Mäntyjärvi, J. (2026). Data-constrained personalization of cognitive state detection with feature-based and foundation models. *User Modeling and User-Adapted Interaction*, 36(2), 9. <https://doi.org/10.1007/s11257-026-09444-w>
- Tierney, M., Adeyemi, E. O., & Bommer, S. (2025). A Systematic Literature Review: Cognitive Workload Assessment in Human Factors Research. *Advances in Human-Computer Interaction*, 2025(1), 9313239. <https://doi.org/10.1155/ahci/9313239>
- Tishby, N., & Zaslavsky, N. (2015). Deep learning and the information bottleneck principle. 2015 IEEE Information Theory Workshop (ITW), 1–5. <https://doi.org/10.1109/ITW.2015.7133169>
- Vapnik, V. N. (2013). *The Nature of Statistical Learning Theory* (2nd ed. 2000). Springer Science & Business Media.
- Veltman, J. A., & Gaillard, A. W. K. (1998). Physiological workload reactions to increasing levels of task difficulty. *Ergonomics*, 41(5), 656–669. <https://doi.org/10.1080/001401398186829>
- Waite, S., Grigorian, A., Alexander, R. G., Macknik, S. L., Carrasco, M., Heeger, D. J., & Martinez-Conde, S. (2019). Analysis of Perceptual Expertise in Radiology – Current Knowledge and a New Perspective. *Frontiers in Human Neuroscience*, 13, 213. <https://doi.org/10.3389/fnhum.2019.00213>
- Wang, D., Shelhamer, E., Liu, S., Olshausen, B. A., & Darrell, T. (2021). Tent: Fully Test-Time Adaptation by Entropy Minimization. 9th International Conference on Learning Representations, ICLR 2021. International Conference on Learning Representations. <https://openreview.net/forum?id=uXl3bZLkr3c>
- Wang, S., Wang, J., Xi, H., Zhang, B., Zhang, L., & Wei, H. (2023). Optimization-Free Test-Time Adaptation for Cross-Person Activity Recognition. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 7(4), 1–27. <https://doi.org/10.1145/3631450>
- Wei, Z., Zhuang, D., Wanyan, X., Liu, C., & Zhuang, H. (2014). A model for discrimination and prediction of mental workload of aircraft cockpit display interface. *Chinese Journal of Aeronautics*, 27(5), 1070–1077. <https://doi.org/10.1016/j.cja.2014.09.002>
- Wickens, C. D. (2002). Multiple resources and performance prediction. *Theoretical Issues in Ergonomics Science*, 3(2), 159–177. <https://doi.org/10.1080/14639220210123806>
- Wilson, G. F. (2002). An Analysis of Mental Workload in Pilots During Flight Using Multiple Psychophysiological Measures. *The International Journal of Aviation Psychology*, 12(1), 3–18. https://doi.org/10.1207/S15327108IJAP1201_2
- Wilson, G. F., & Russell, C. A. (2003). Operator Functional State Classification Using Multiple Psychophysiological Features in an Air Traffic Control Task. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 45(3), 381–389. <https://doi.org/10.1518/hfes.45.3.381.27252>
- Yang, D., Huang, S., Kuang, H., Du, Y., & Zhang, L. (2022). Disentangled Representation Learning for Multimodal Emotion Recognition. *Proceedings of the 30th ACM International Conference on Multimedia*, 1642–1651. <https://doi.org/10.1145/3503161.3547754>
- Yang, H., Wu, J., Hu, Z., & Lv, C. (2024). Real-Time Driver Cognitive Workload Recognition: Attention-Enabled Learning With Multimodal Information Fusion. *IEEE Transactions on Industrial Electronics*, 71(5), 4999–5009. <https://doi.org/10.1109/TIE.2023.3288182>
- Young, M. S., Brookhuis, K. A., Wickens, C. D., & Hancock, P. A. (2015). State of science: Mental workload in ergonomics. *Ergonomics*, 58(1), 1–17. <https://doi.org/10.1080/00140139.2014.956151>
- Young, M. S., & Stanton, N. A. (2002). Malleable Attentional Resources Theory: A New Explanation for the Effects of Mental Underload on Performance. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 44(3), 365–375. <https://doi.org/10.1518/0018720024497709>
- Zhang, C., Bengio, S., Hardt, M., Recht, B., & Vinyals, O. (2017). Understanding Deep Learning Requires Rethinking Generalization. *International Conference on Learning Representations*. International Conference on Learning Representations.
- Zhang, C., Jiang, C., Xie, Y., Cao, S., Yuan, J., Liu, C., Cao, W., & Li, Y. (2024). Assessing Pilot Workload During Takeoff and Climb Under Different Weather Conditions: A fNIRS-Based Modeling Using Deep Learning Algorithms. *IEEE Transactions on Aerospace and Electronic Systems*, 61(2), 1705–1724. <https://doi.org/10.1109/TAES.2024.3458954>

- Zhao, G., Liu, Y.-J., & Shi, Y. (2018). Real-Time Assessment of the Cross-Task Mental Workload Using Physiological Measures During Anomaly Detection. *IEEE Transactions on Human-Machine Systems*, 48(2), 149–160. <https://doi.org/10.1109/THMS.2018.2803025>
- Zhou, T., Cha, J. S., Gonzalez, G., Wachs, J. P., Sundaram, C. P., & Yu, D. (2020). Multimodal Physiological Signals for Workload Prediction in Robot-assisted Surgery. *ACM Transactions on Human-Robot Interaction*, 9(2), 1–26. <https://doi.org/10.1145/3368589>
- Zhou, Y., Huang, S., Xu, Z., Wang, P., Wu, X., & Zhang, D. (2022). Cognitive Workload Recognition Using EEG Signals and Machine Learning: A Review. *IEEE Transactions on Cognitive and Developmental Systems*, 14(3), 799–818. <https://doi.org/10.1109/TCDS.2021.3090217>
- Zhou, Y., Wang, P., Gong, P., Liu, Y., Wen, X., Wu, X., & Zhang, D. (2023). Deep Domain Adaptation for EEG-Based Cross-Subject Cognitive Workload Recognition. In M. Tanveer, S. Agarwal, S. Ozawa, A. Ekbal, & A. Jatowt (Eds.), *Neural Information Processing* (pp. 231–242). Springer Nature. https://doi.org/10.1007/978-981-99-1642-9_20
- Zubair, M., Woo, S., Lim, S., & Yoon, C. (2024). Deep Representation Learning for Multimodal Emotion Recognition Using Physiological Signals. *IEEE Access*, 12, 106605–106617. <https://doi.org/10.1109/ACCESS.2024.3436556>